

BFS Traversal

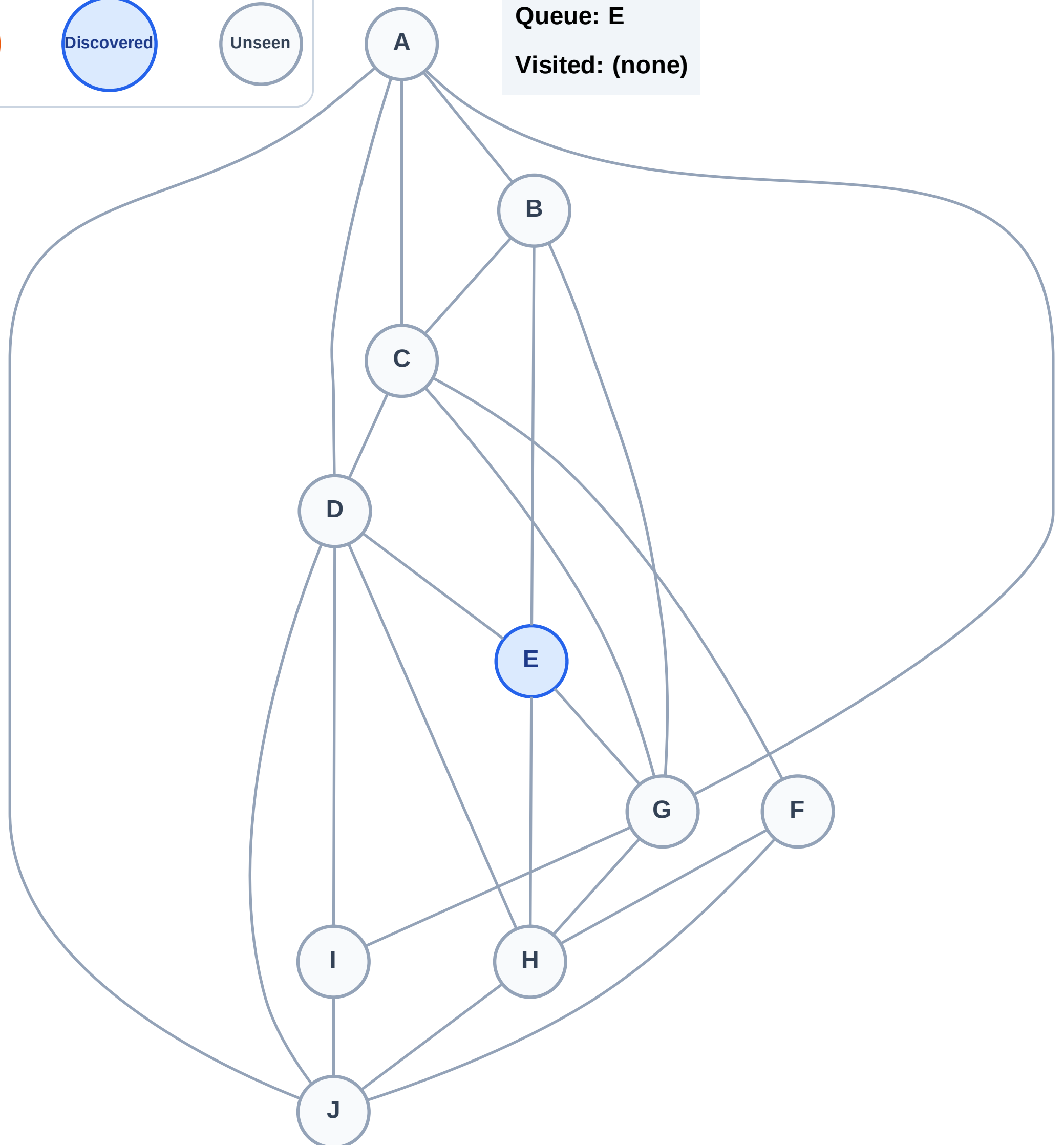
Step 1: Start at E

Legend



Queue: E

Visited: (none)



BFS Traversal

Step 2: Process node E

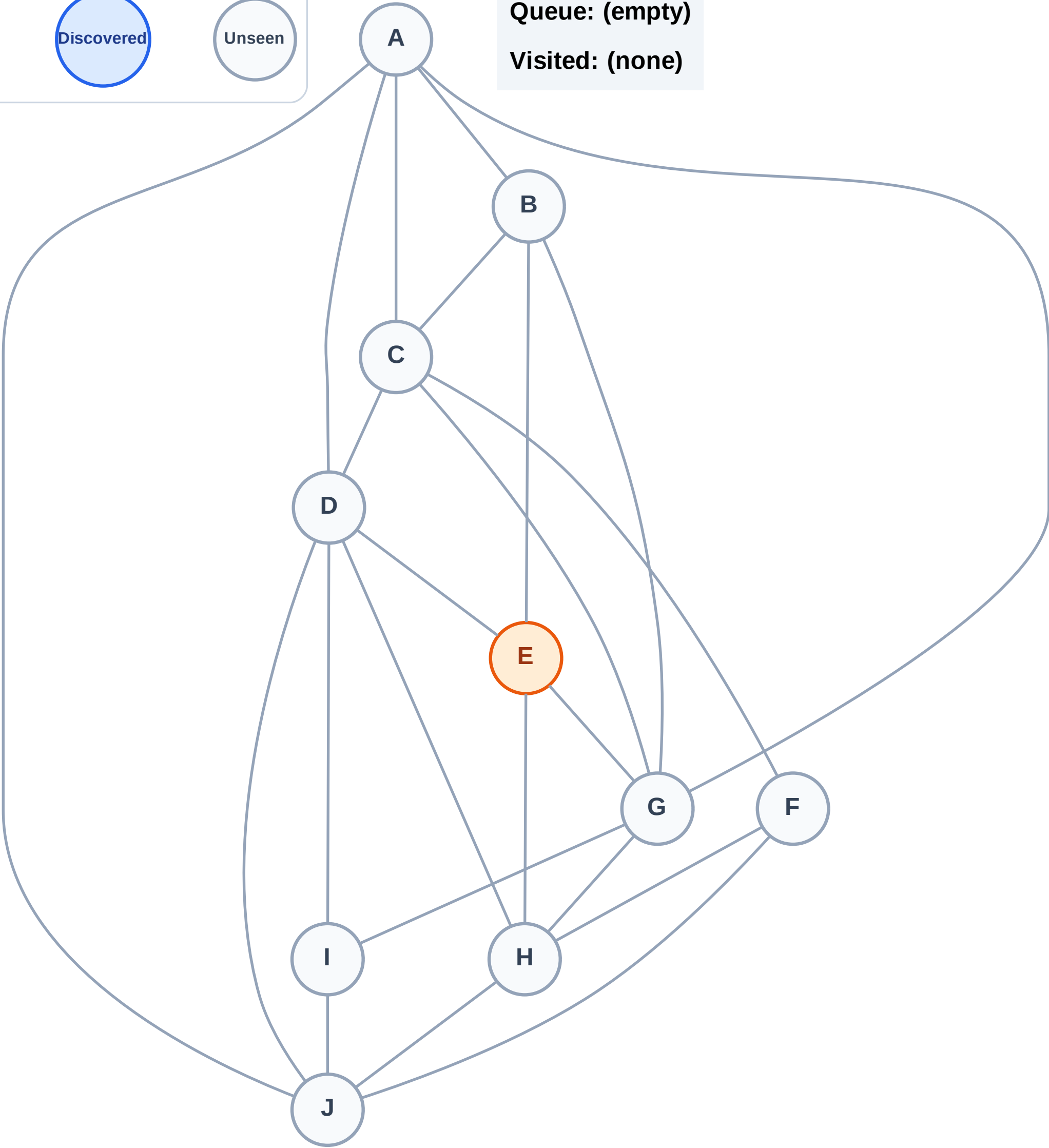
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



BFS Traversal

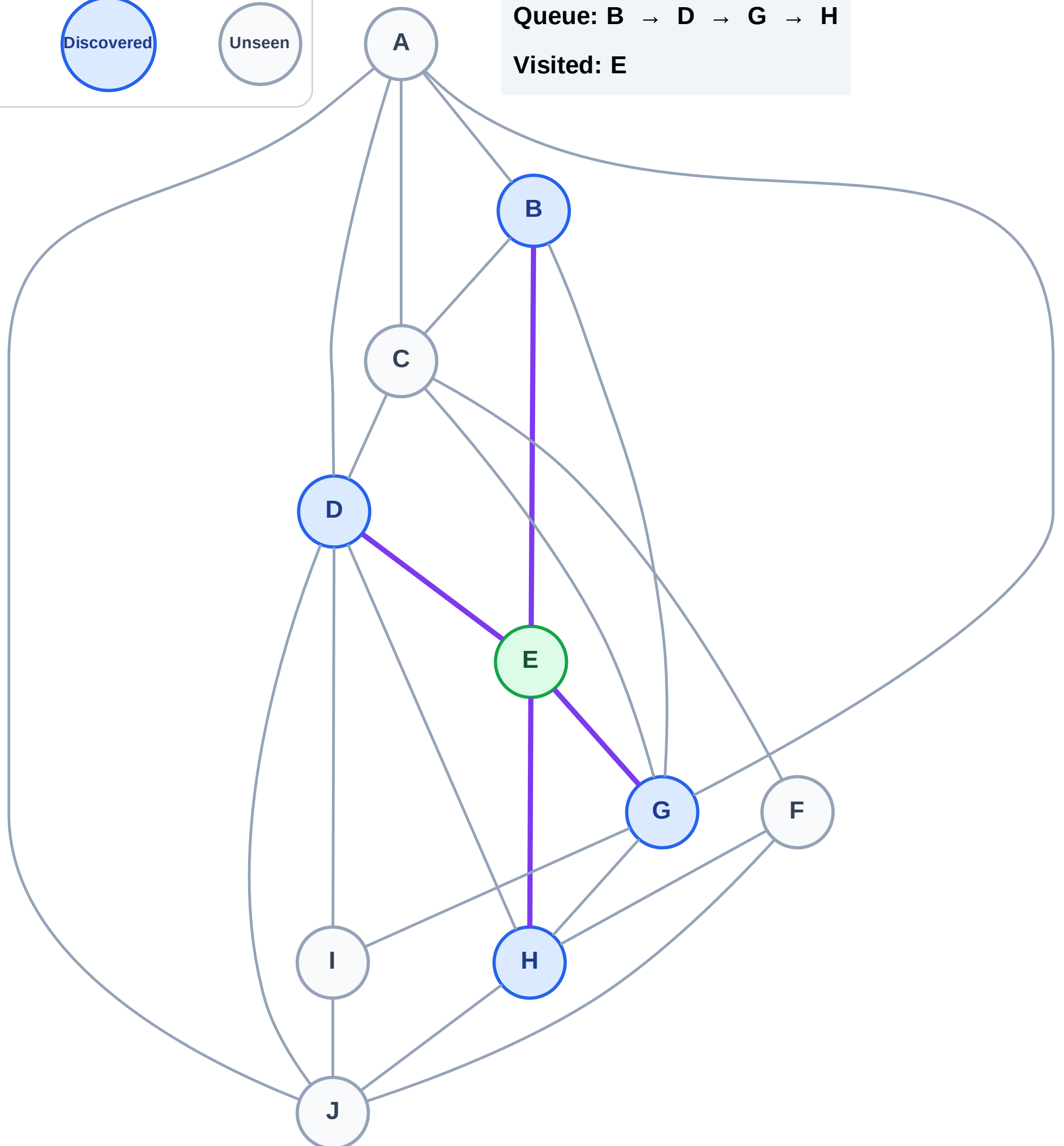
Step 3: Discover B, D, G, H from E

Legend



Queue: B → D → G → H

Visited: E



BFS Traversal

Step 4: Process node B

Legend

Complete

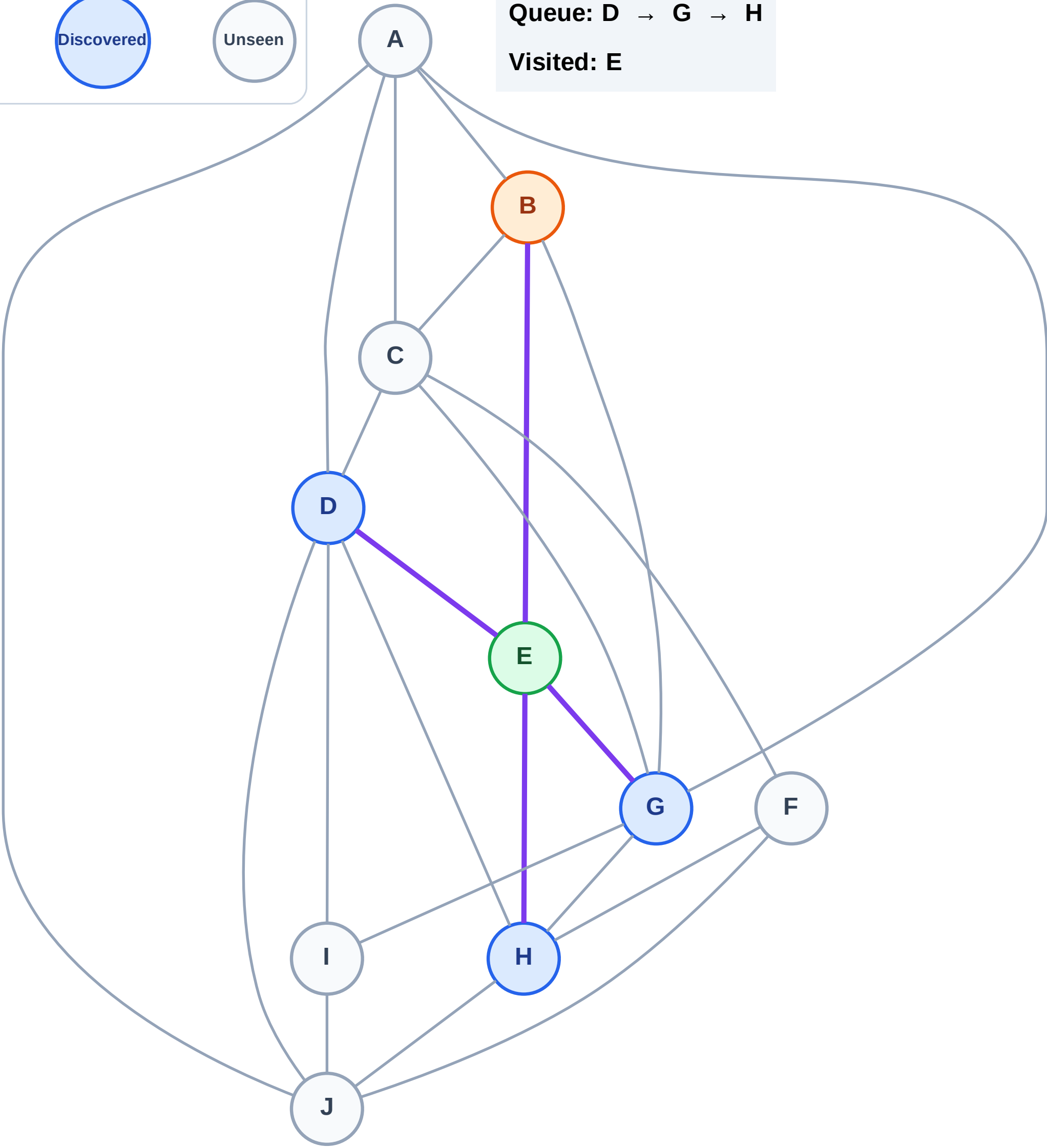
Processing

Discovered

Unseen

Queue: D → G → H

Visited: E



Step 5: Discover A, C from B

Step 5: Discover A, C from B



BFS Traversal

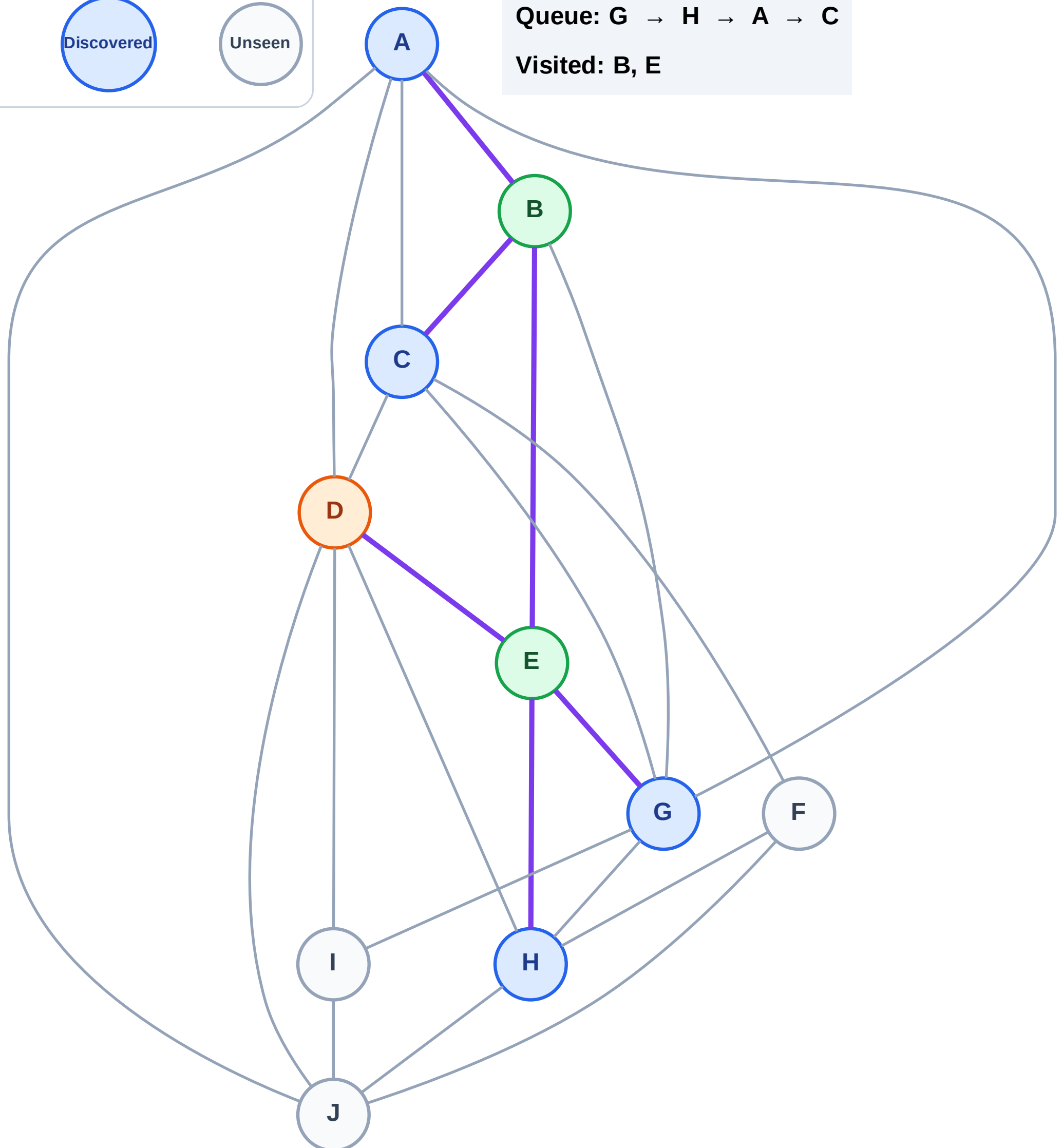
Step 6: Process node D

Legend



Queue: G → H → A → C

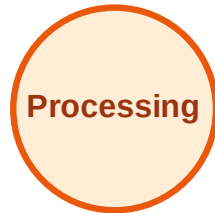
Visited: B, E



BFS Traversal

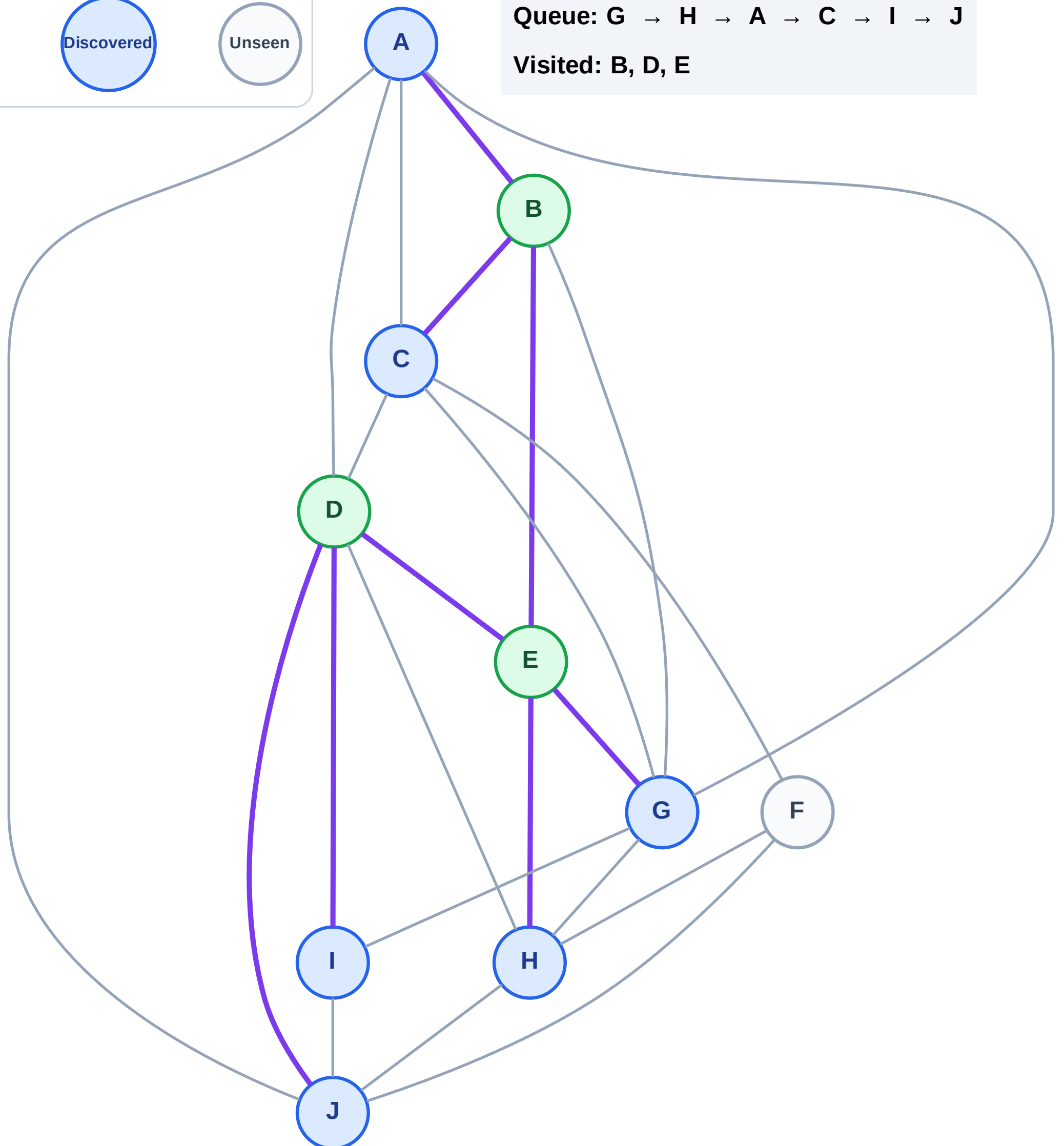
Step 7: Discover I, J from D

Legend



Queue: G → H → A → C → I → J

Visited: B, D, E



BFS Traversal

Step 8: Process node G

Legend

Complete

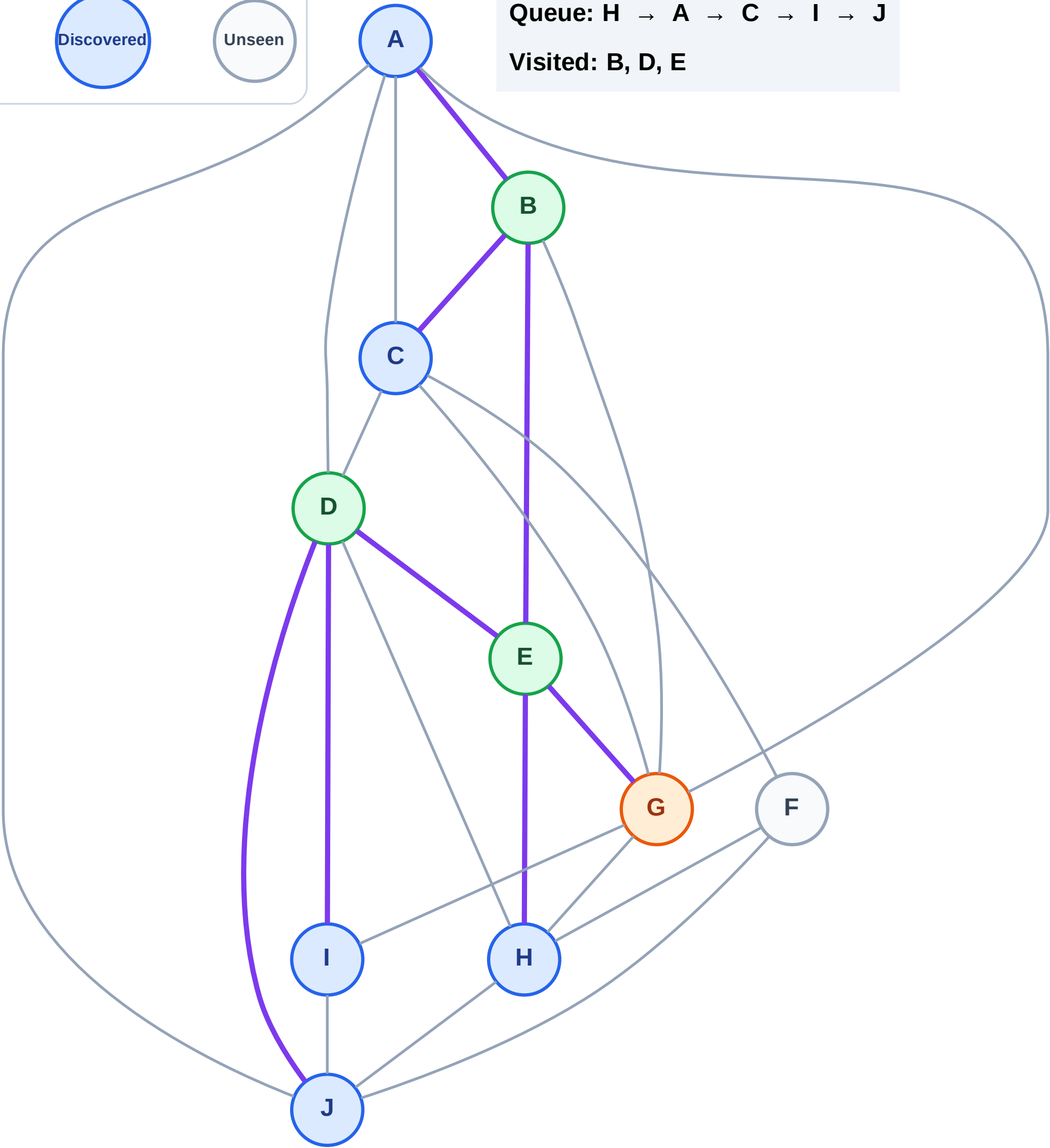
Processing

Discovered

Unseen

Queue: H → A → C → I → J

Visited: B, D, E



BFS Traversal

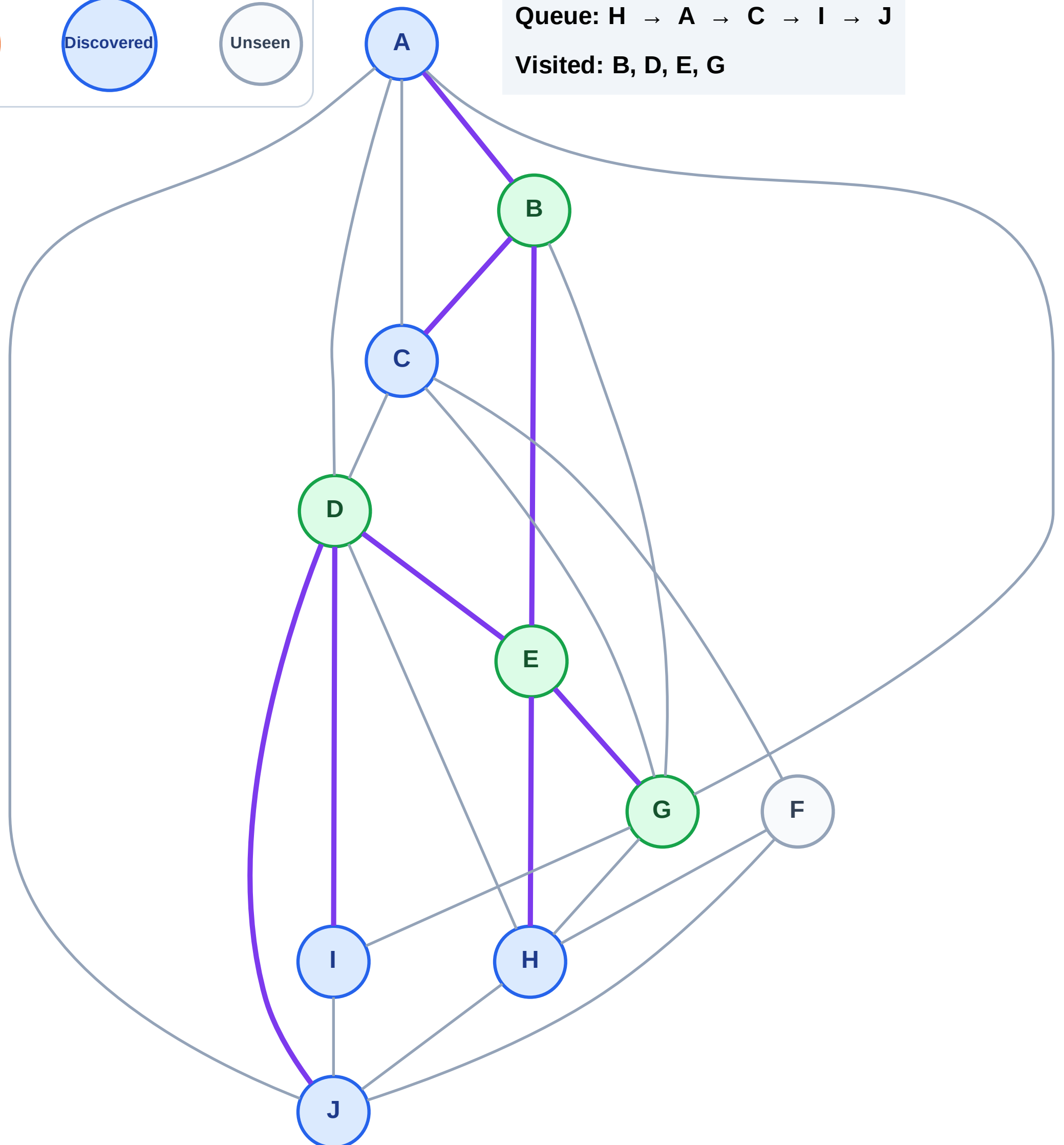
Step 9: No new neighbors from G

Legend



Queue: H → A → C → I → J

Visited: B, D, E, G



BFS Traversal

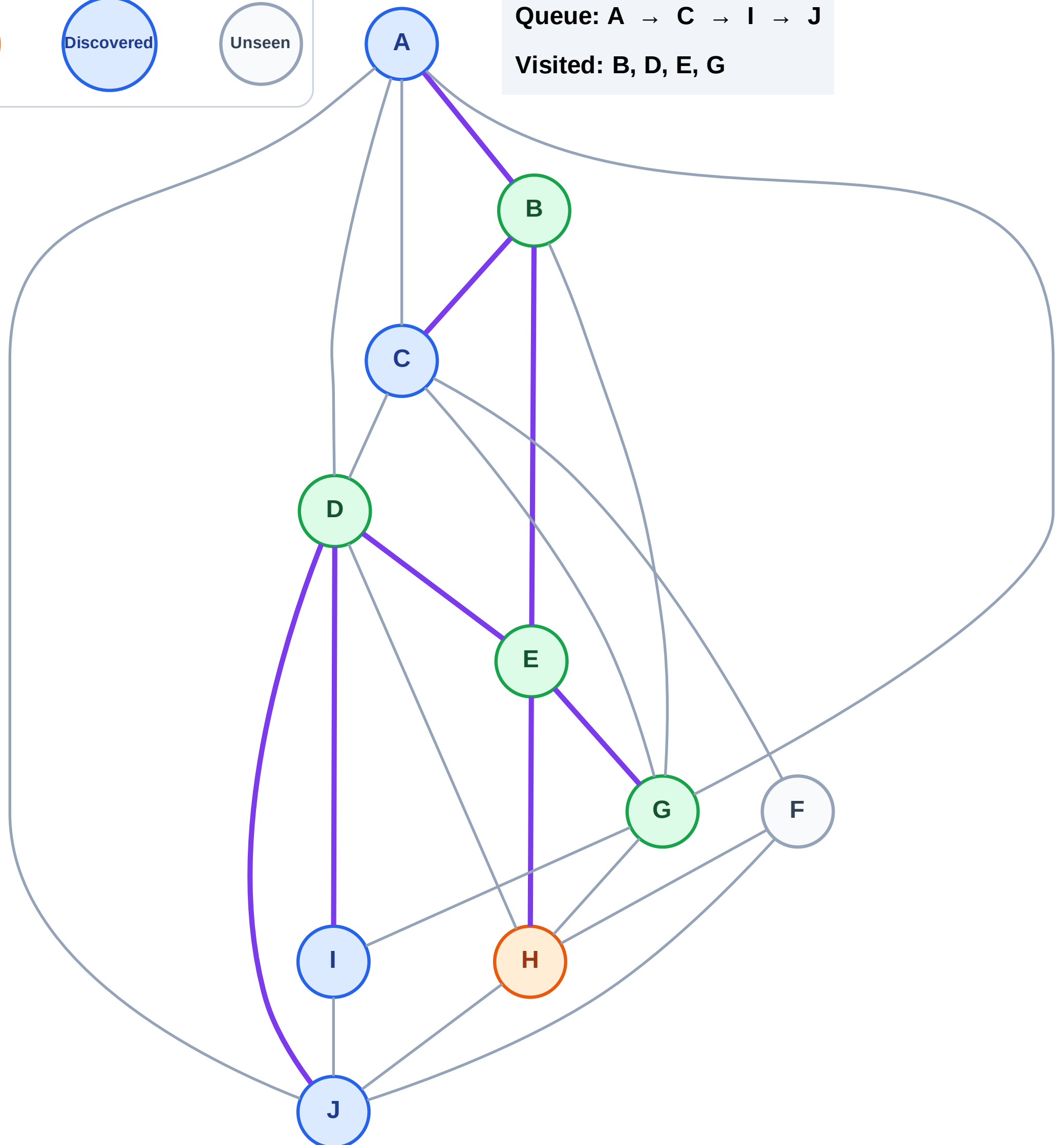
Step 10: Process node H

Legend



Queue: A → C → I → J

Visited: B, D, E, G



BFS Traversal

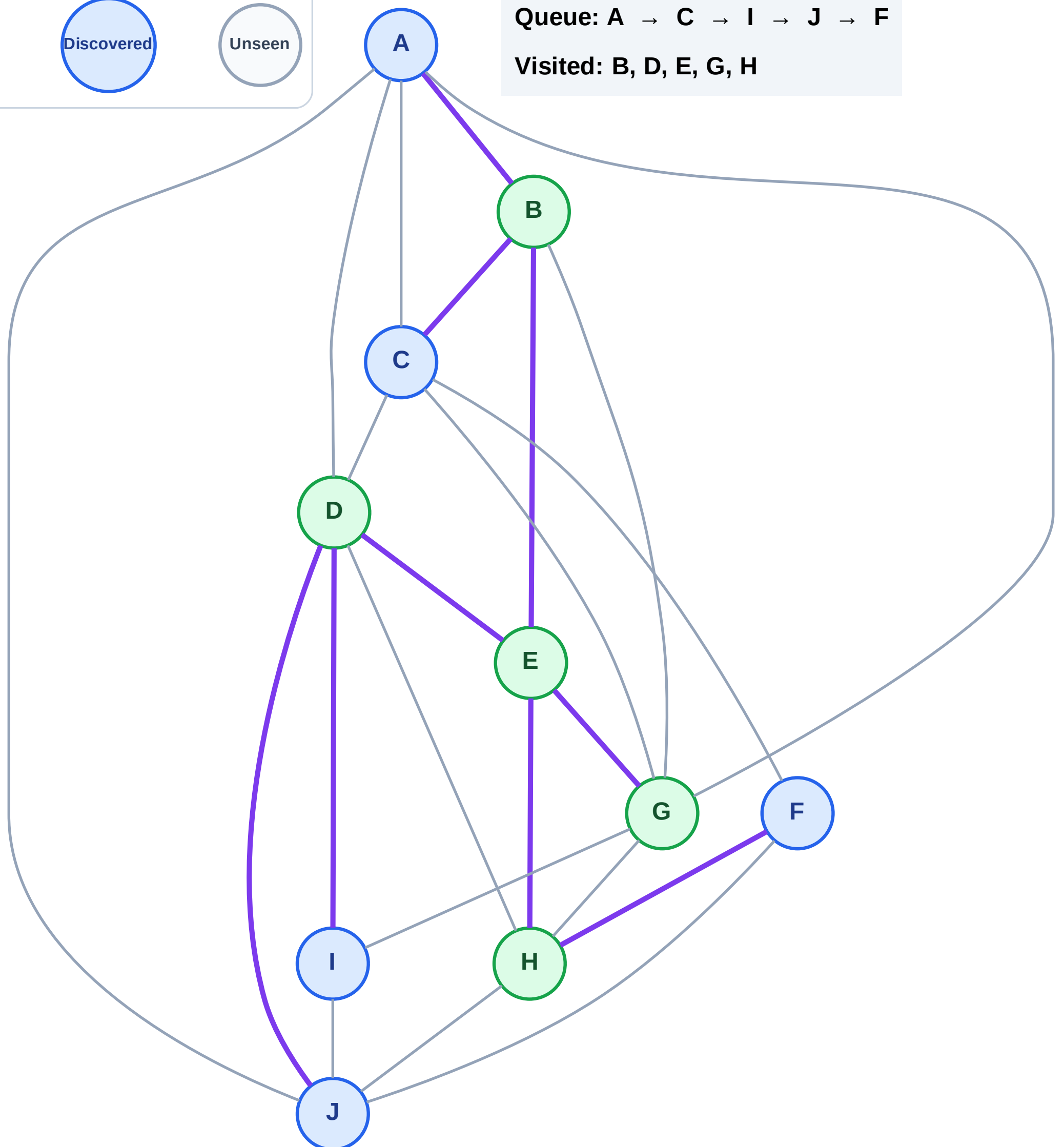
Step 11: Discover F from H

Legend



Queue: A → C → I → J → F

Visited: B, D, E, G, H



BFS Traversal

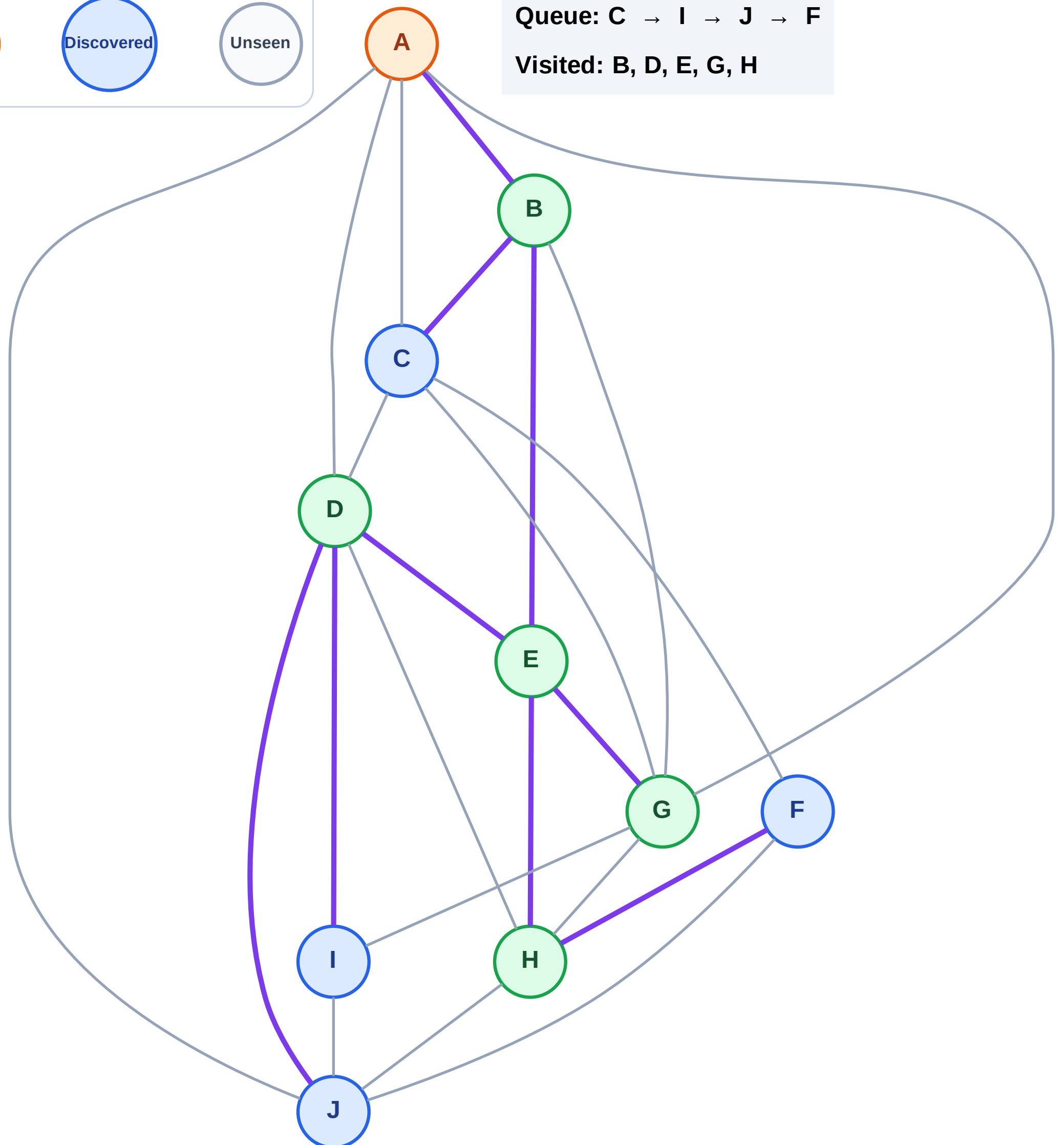
Step 12: Process node A

Legend



Queue: C → I → J → F

Visited: B, D, E, G, H



BFS Traversal

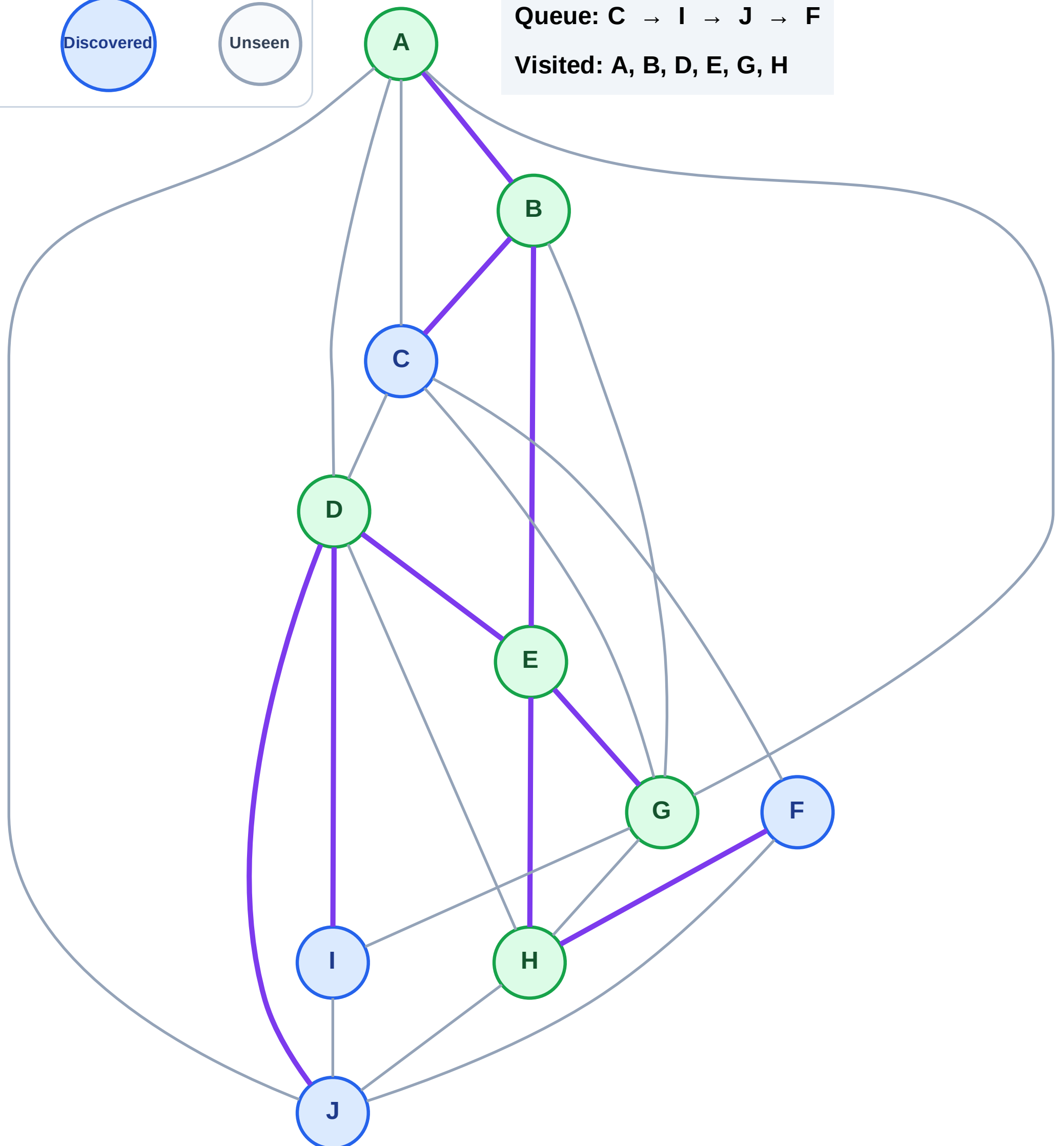
Step 13: No new neighbors from A

Legend



Queue: C → I → J → F

Visited: A, B, D, E, G, H



BFS Traversal

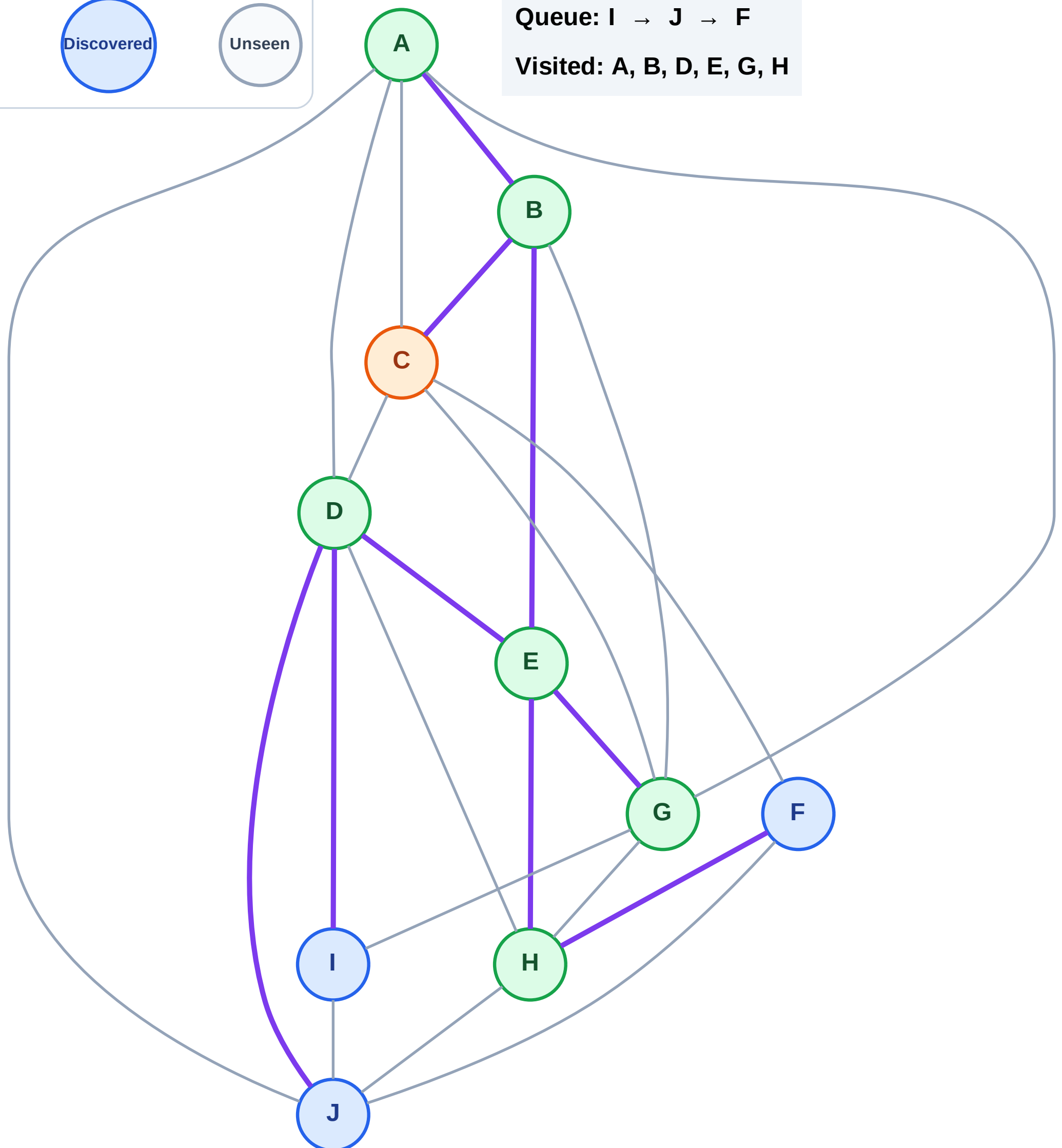
Step 14: Process node C

Legend



Queue: I → J → F

Visited: A, B, D, E, G, H



BFS Traversal

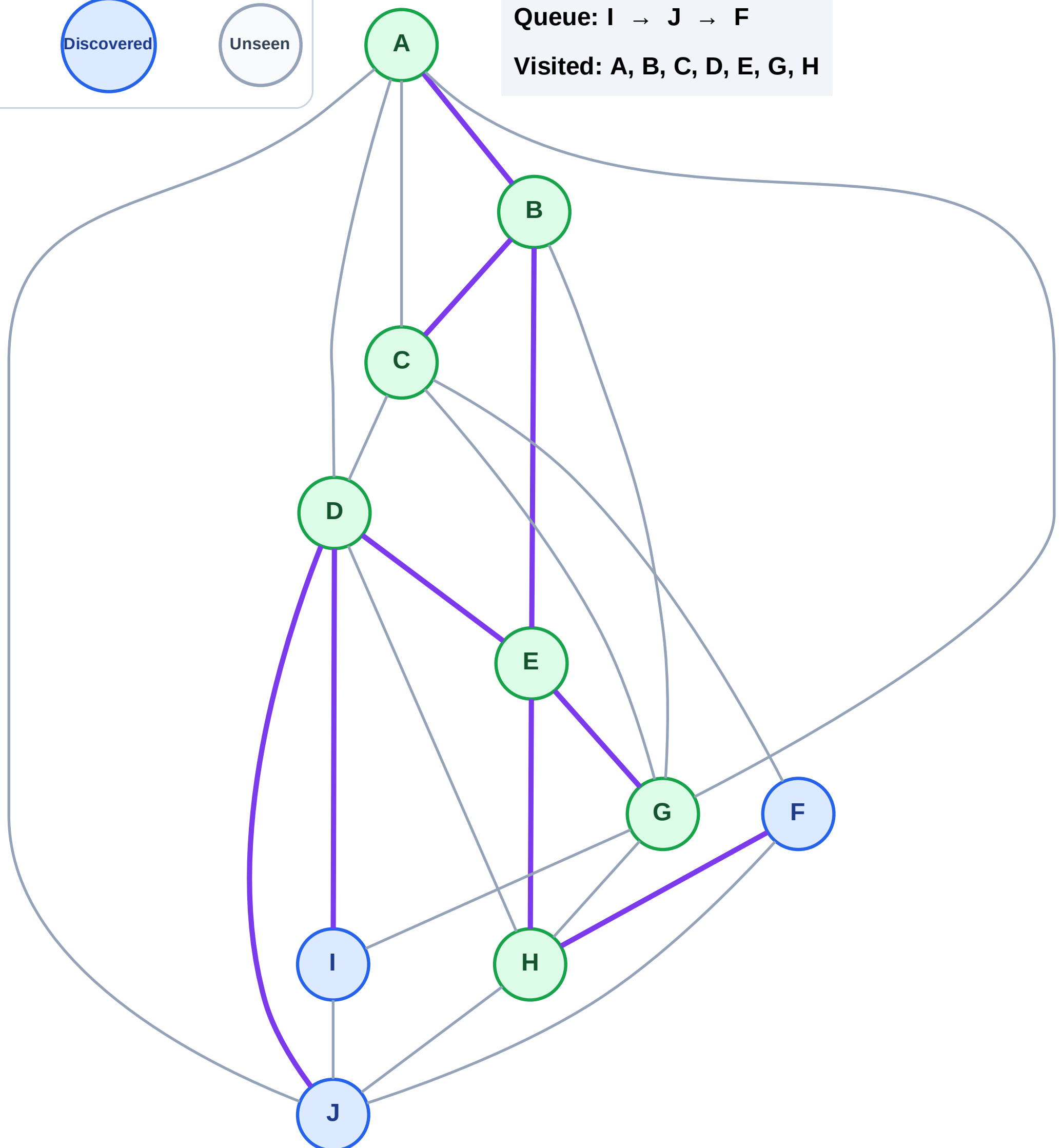
Step 15: No new neighbors from C

Legend



Queue: I → J → F

Visited: A, B, C, D, E, G, H



BFS Traversal

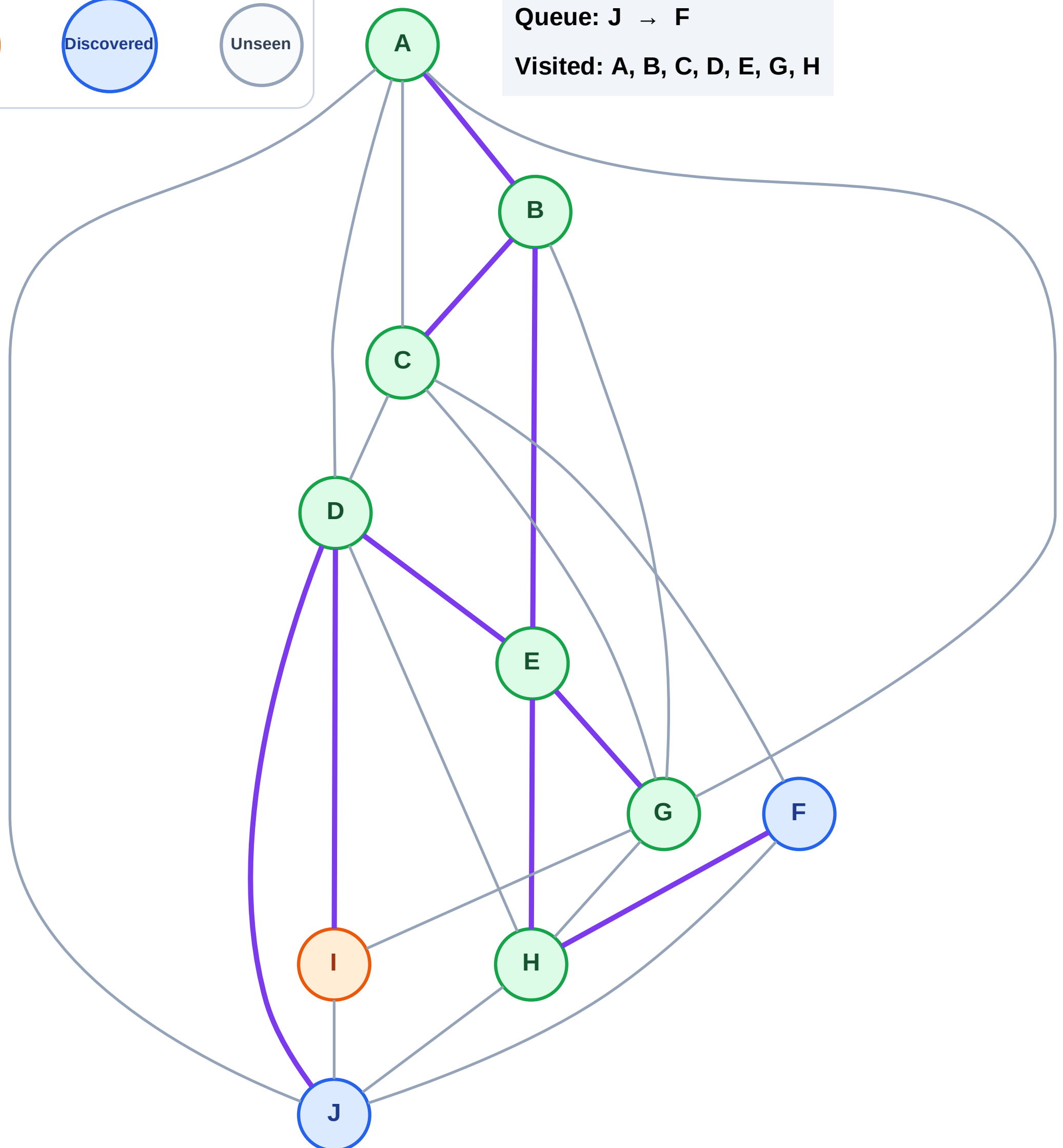
Step 16: Process node I

Legend



Queue: J → F

Visited: A, B, C, D, E, G, H



BFS Traversal

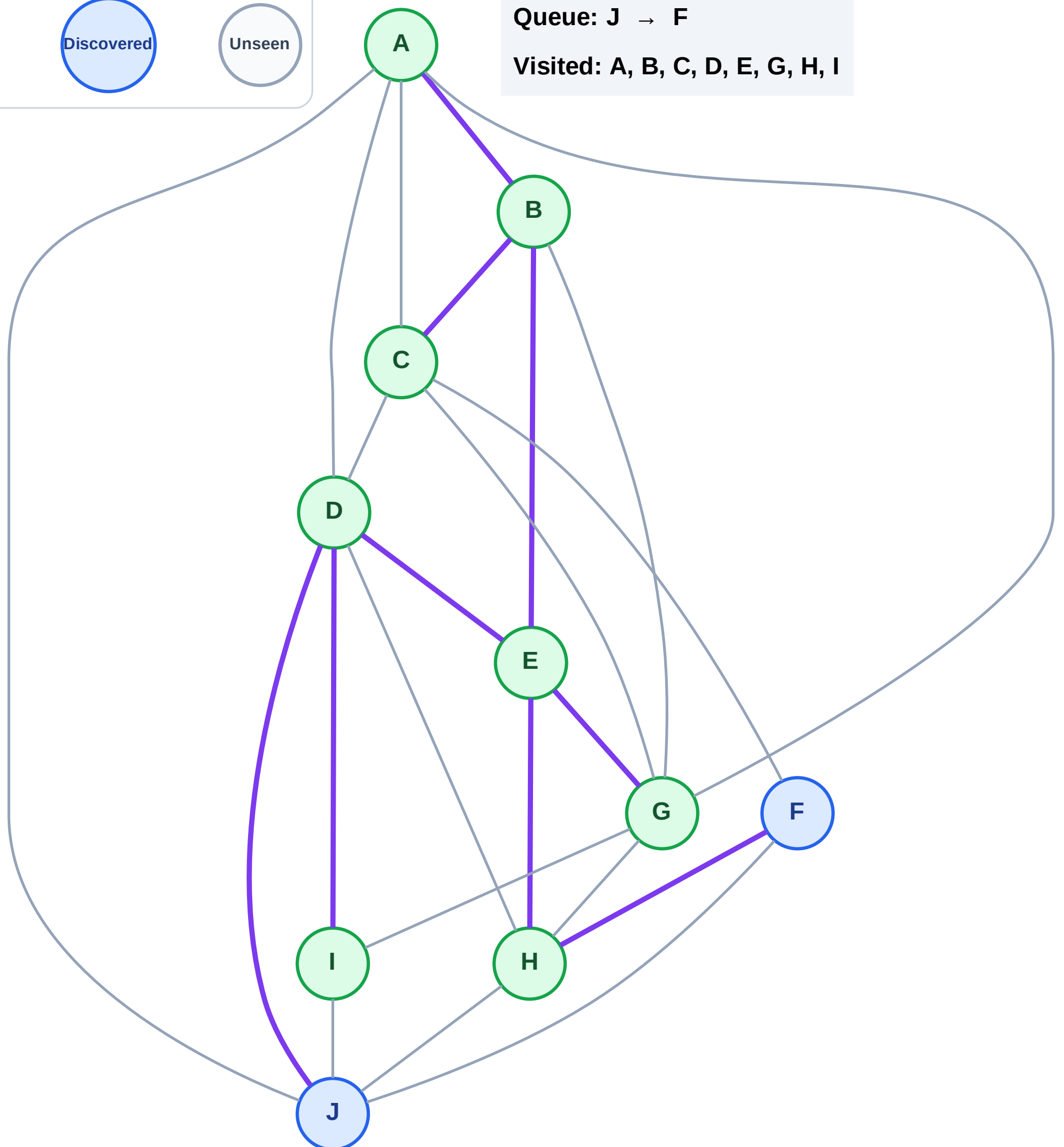
Step 17: No new neighbors from I

Legend



Queue: J → F

Visited: A, B, C, D, E, G, H, I



BFS Traversal

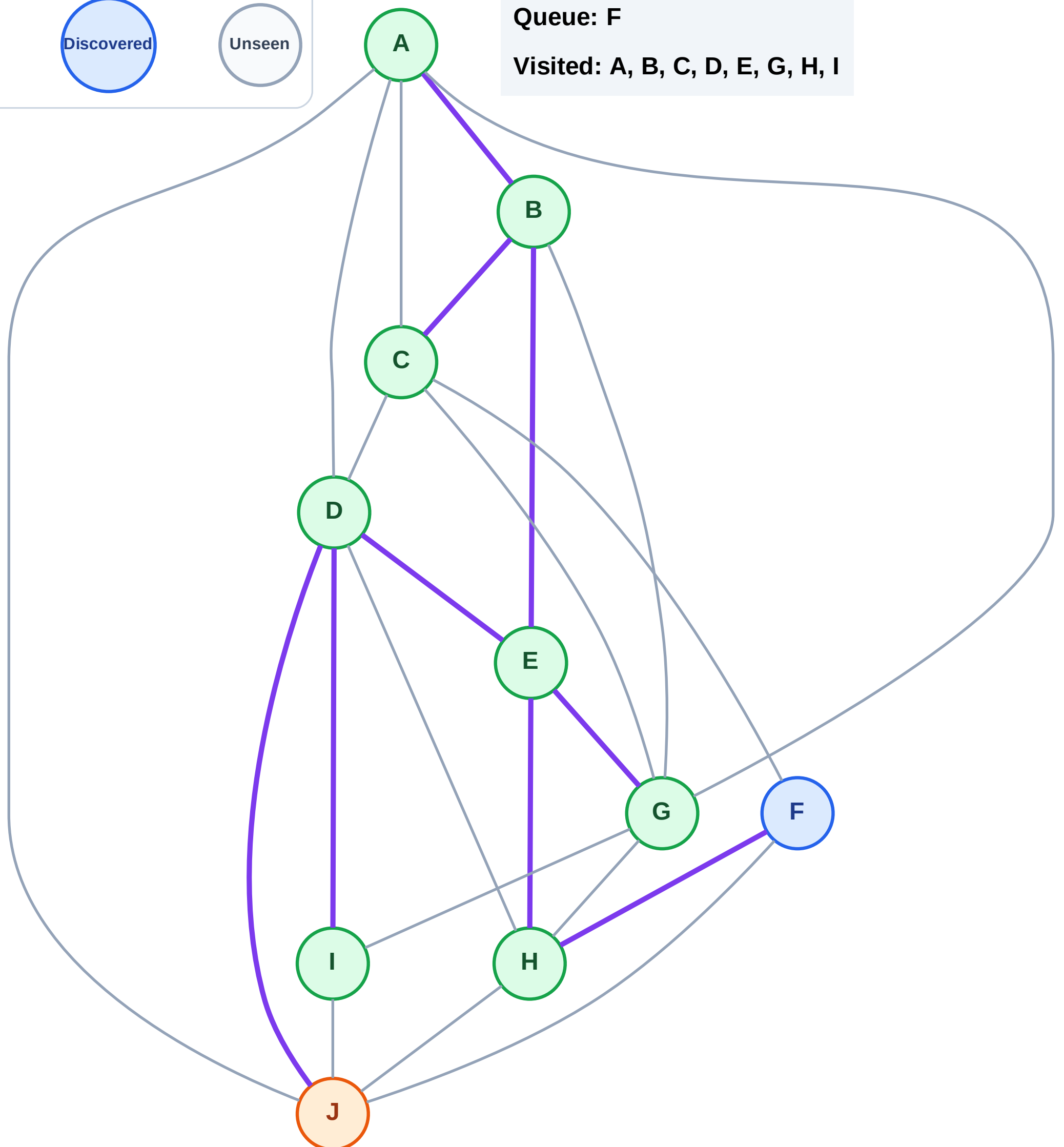
Step 18: Process node J

Legend



Queue: F

Visited: A, B, C, D, E, G, H, I



BFS Traversal

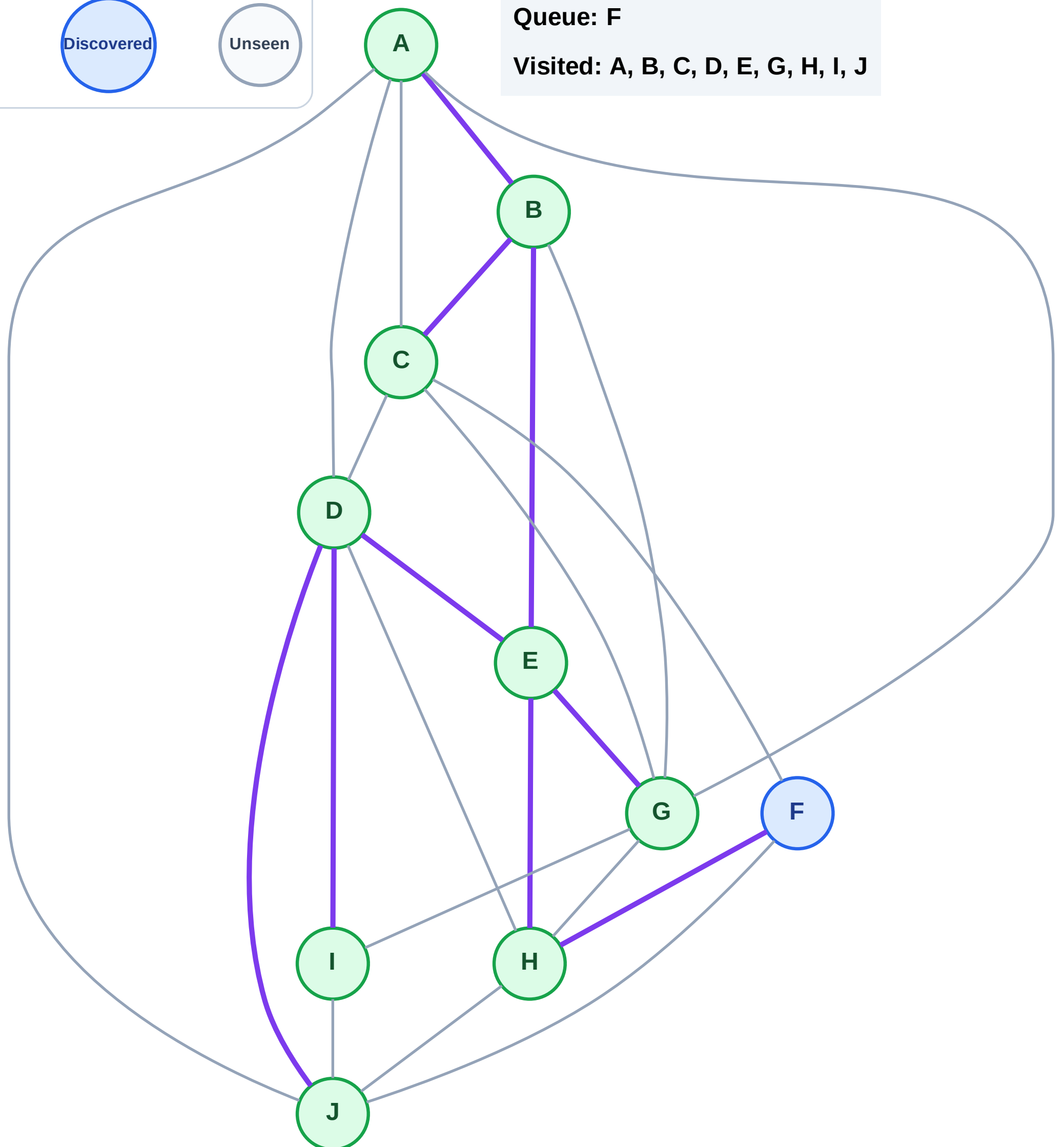
Step 19: No new neighbors from J

Legend



Queue: F

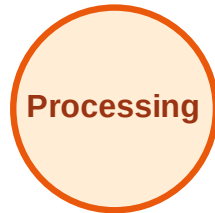
Visited: A, B, C, D, E, G, H, I, J



BFS Traversal

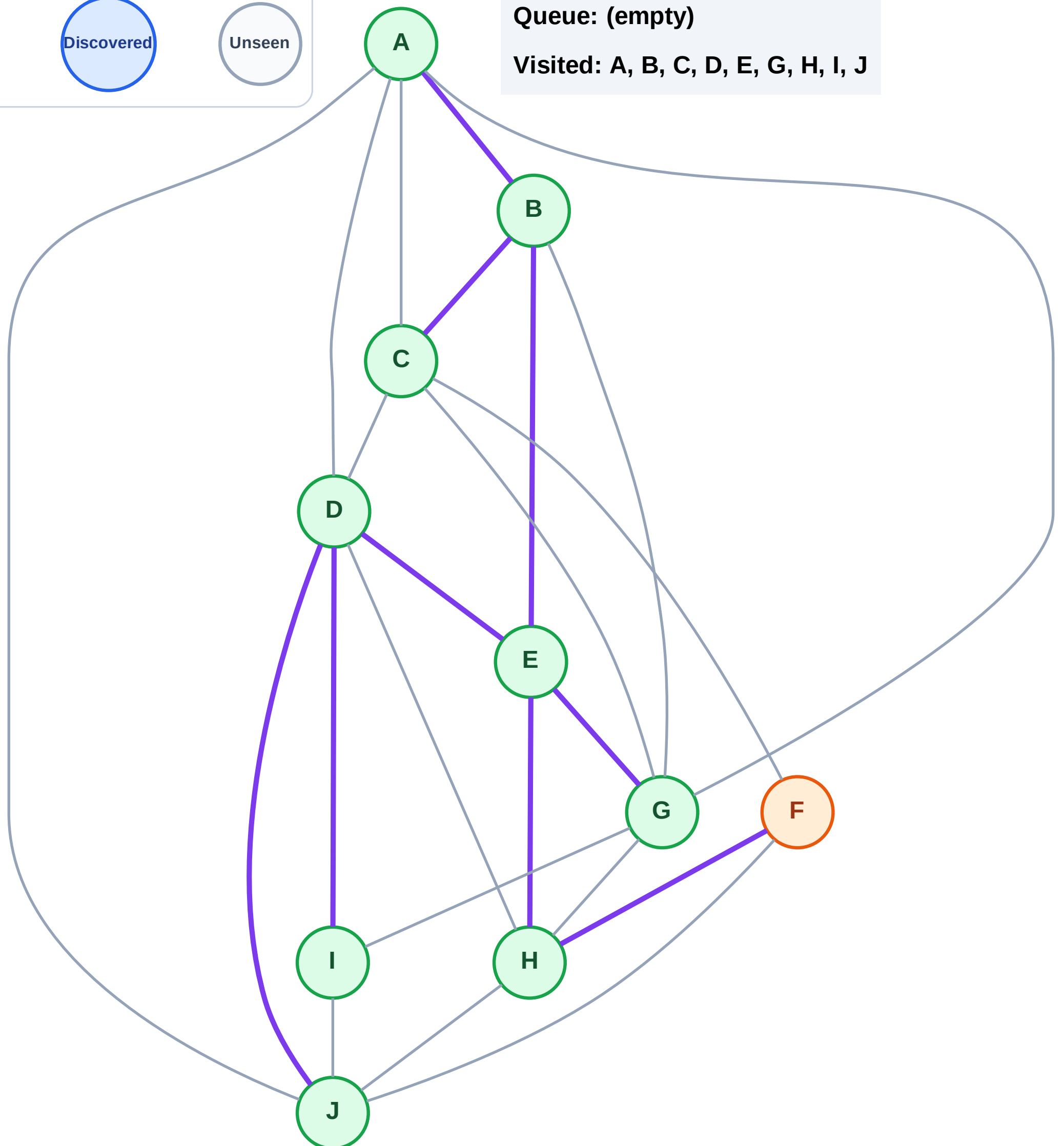
Step 20: Process node F

Legend



Queue: (empty)

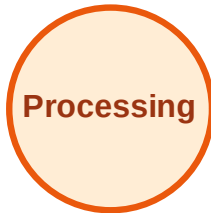
Visited: A, B, C, D, E, G, H, I, J



BFS Traversal

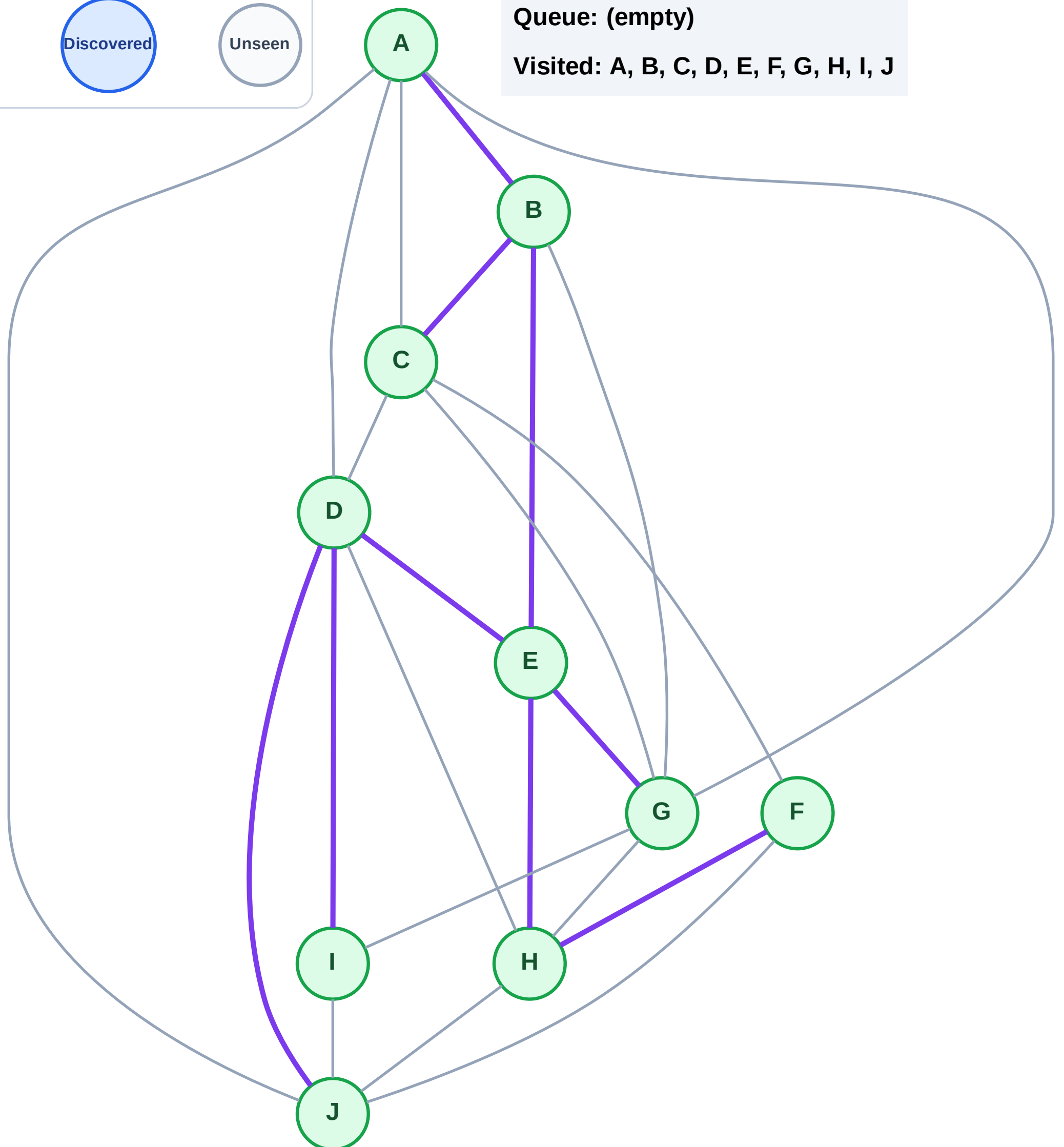
Step 21: No new neighbors from F

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

Step 1: Start at E

Legend

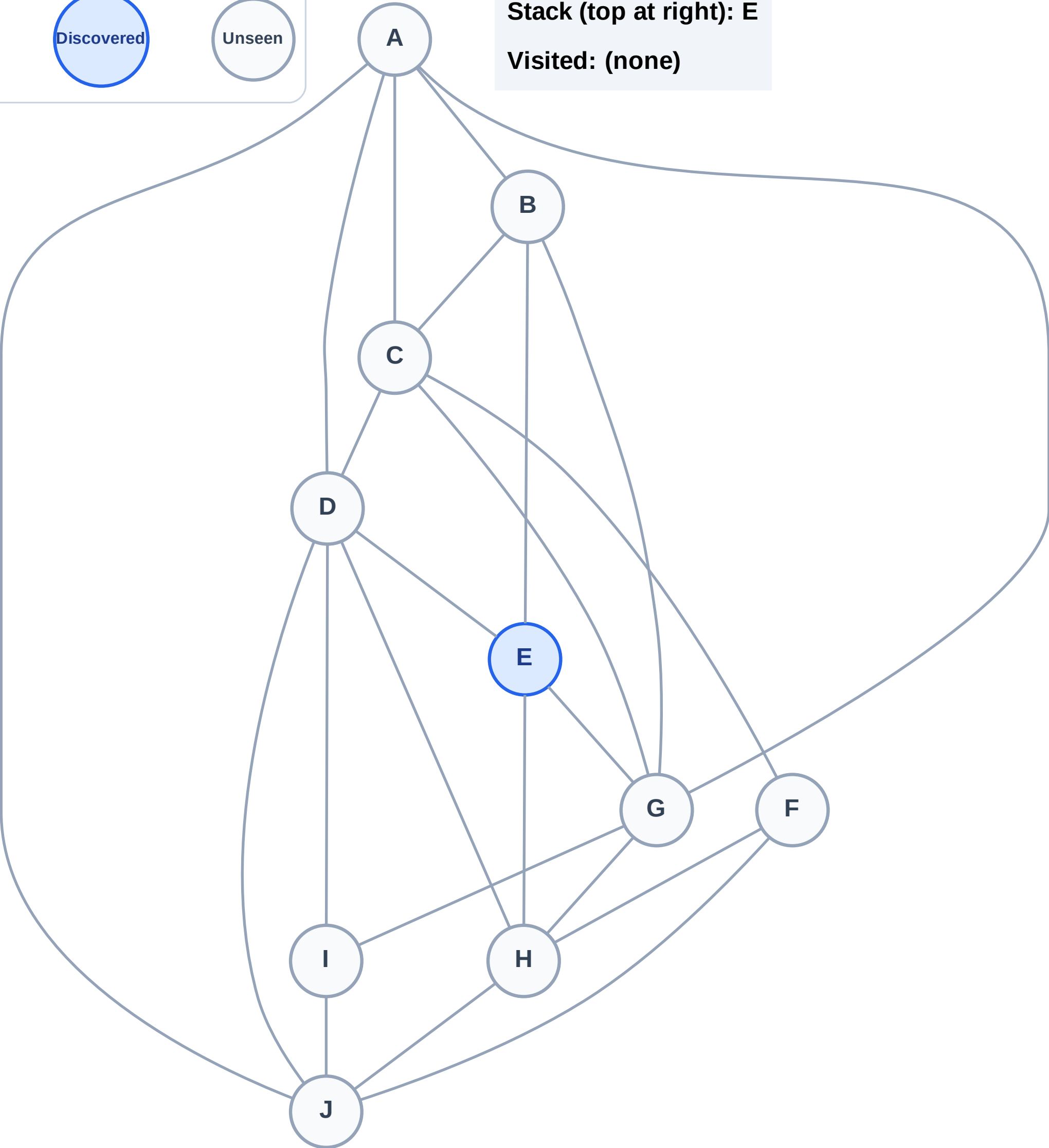
Complete

Processing

Discovered

Unseen

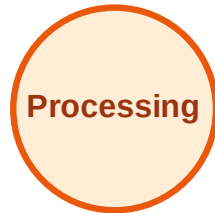
Stack (top at right): E
Visited: (none)



DFS Traversal

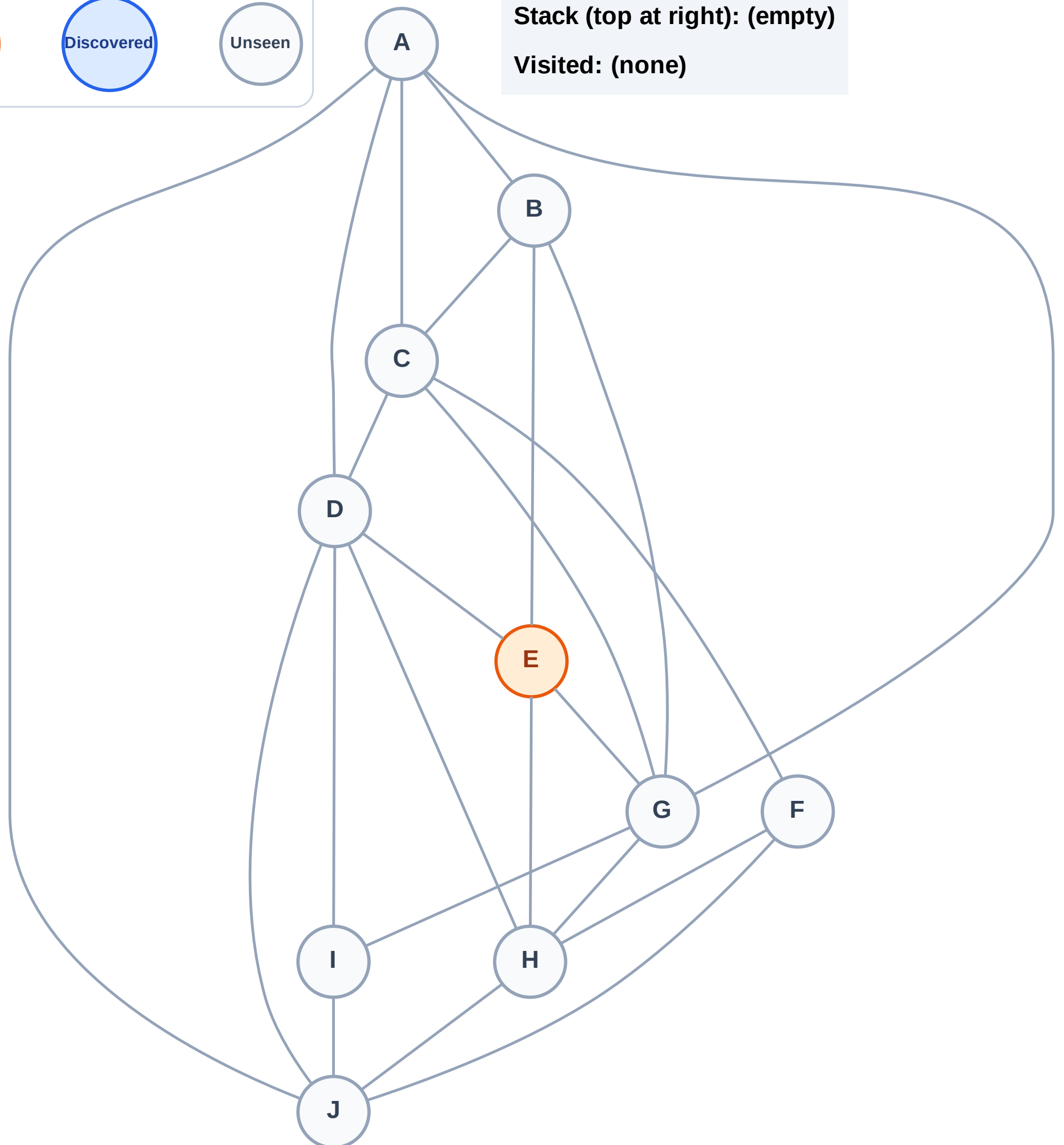
Step 2: Process node E

Legend



Stack (top at right): (empty)

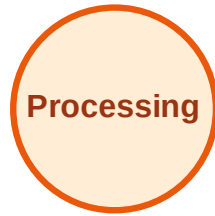
Visited: (none)



DFS Traversal

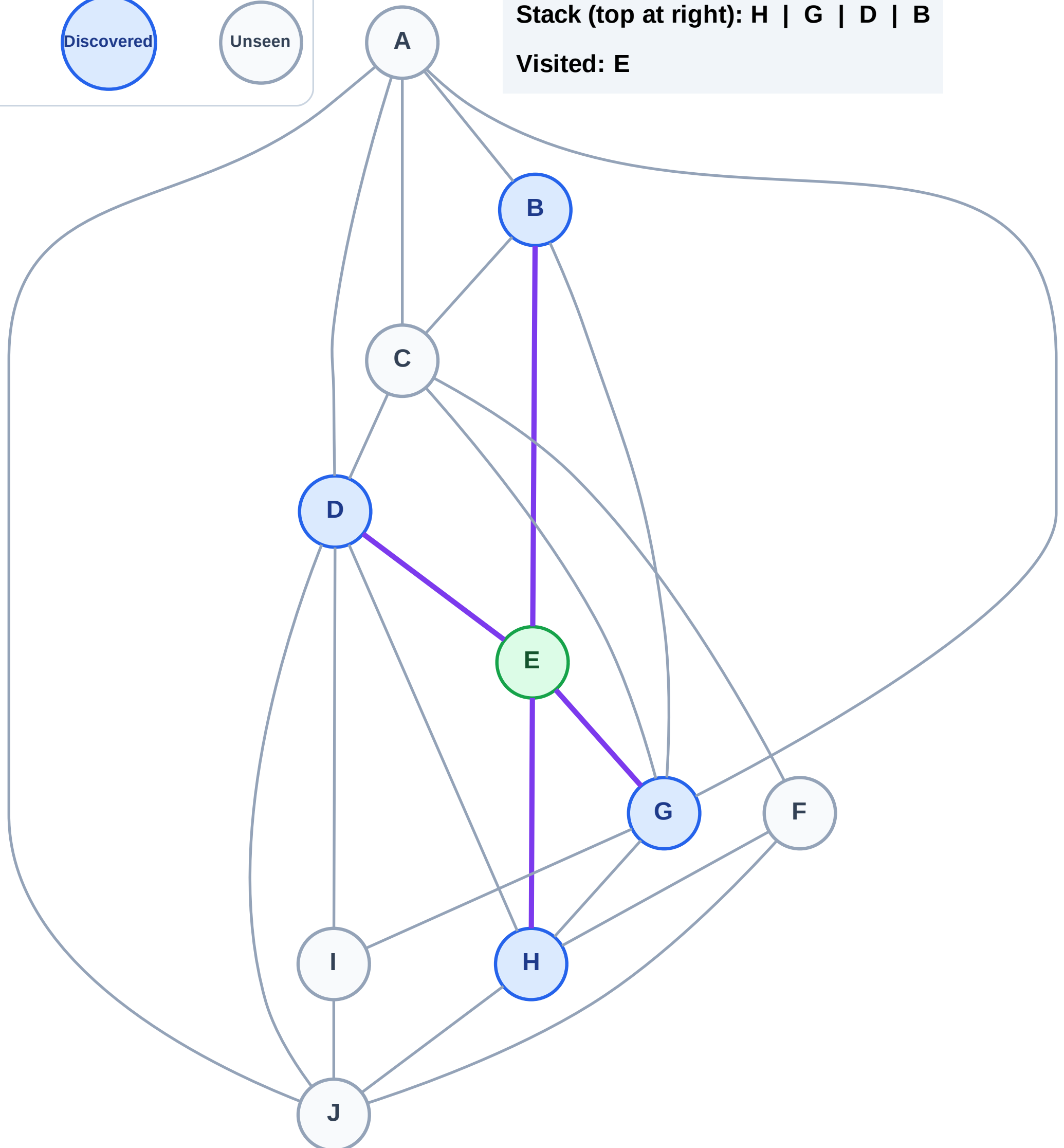
Step 3: Discover H, G, D, B from E

Legend



Stack (top at right): H | G | D | B

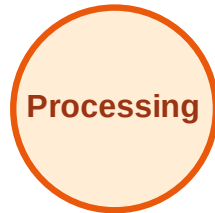
Visited: E



DFS Traversal

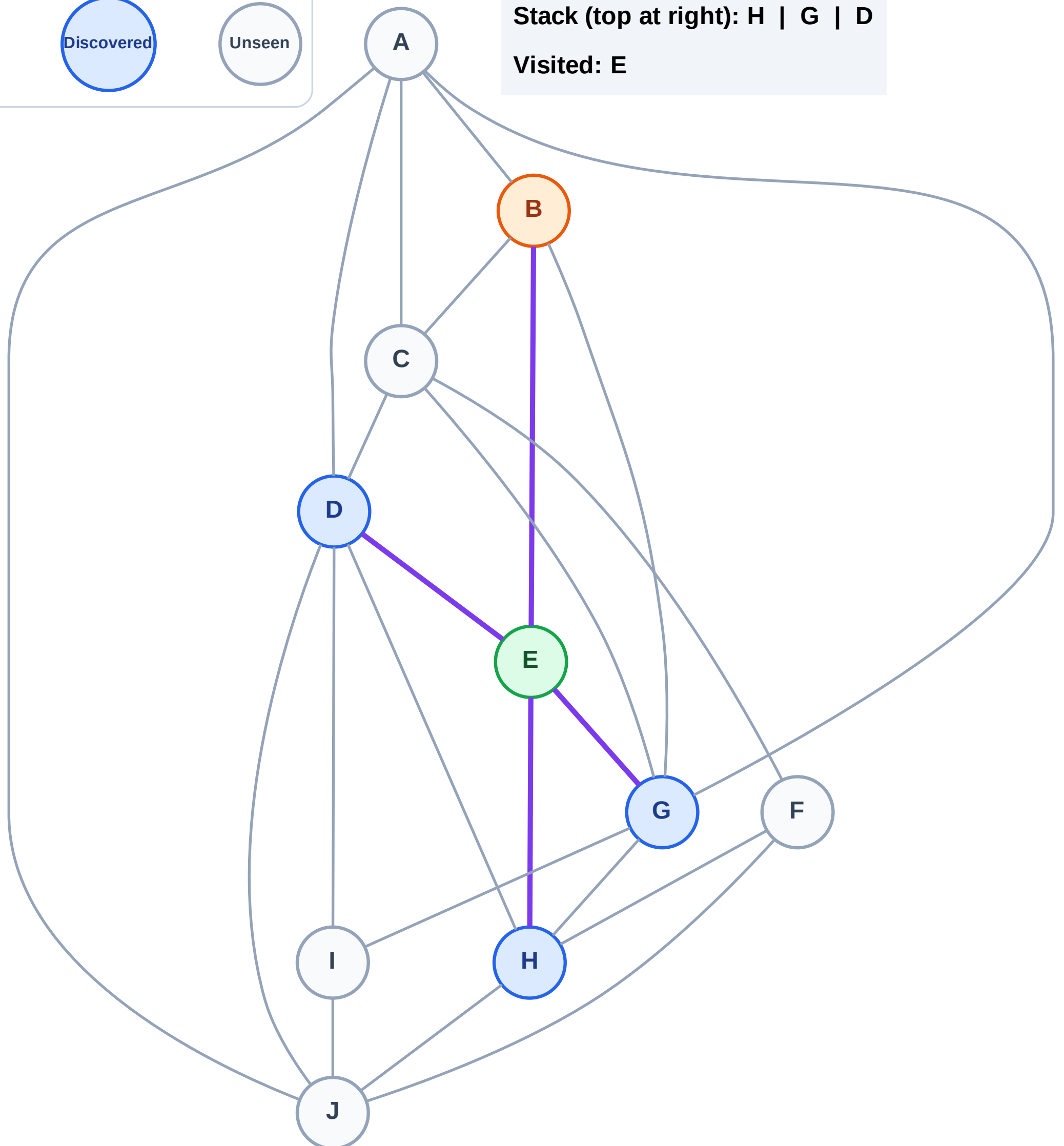
Step 4: Process node B

Legend



Stack (top at right): H | G | D

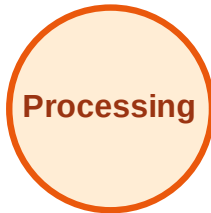
Visited: E



DFS Traversal

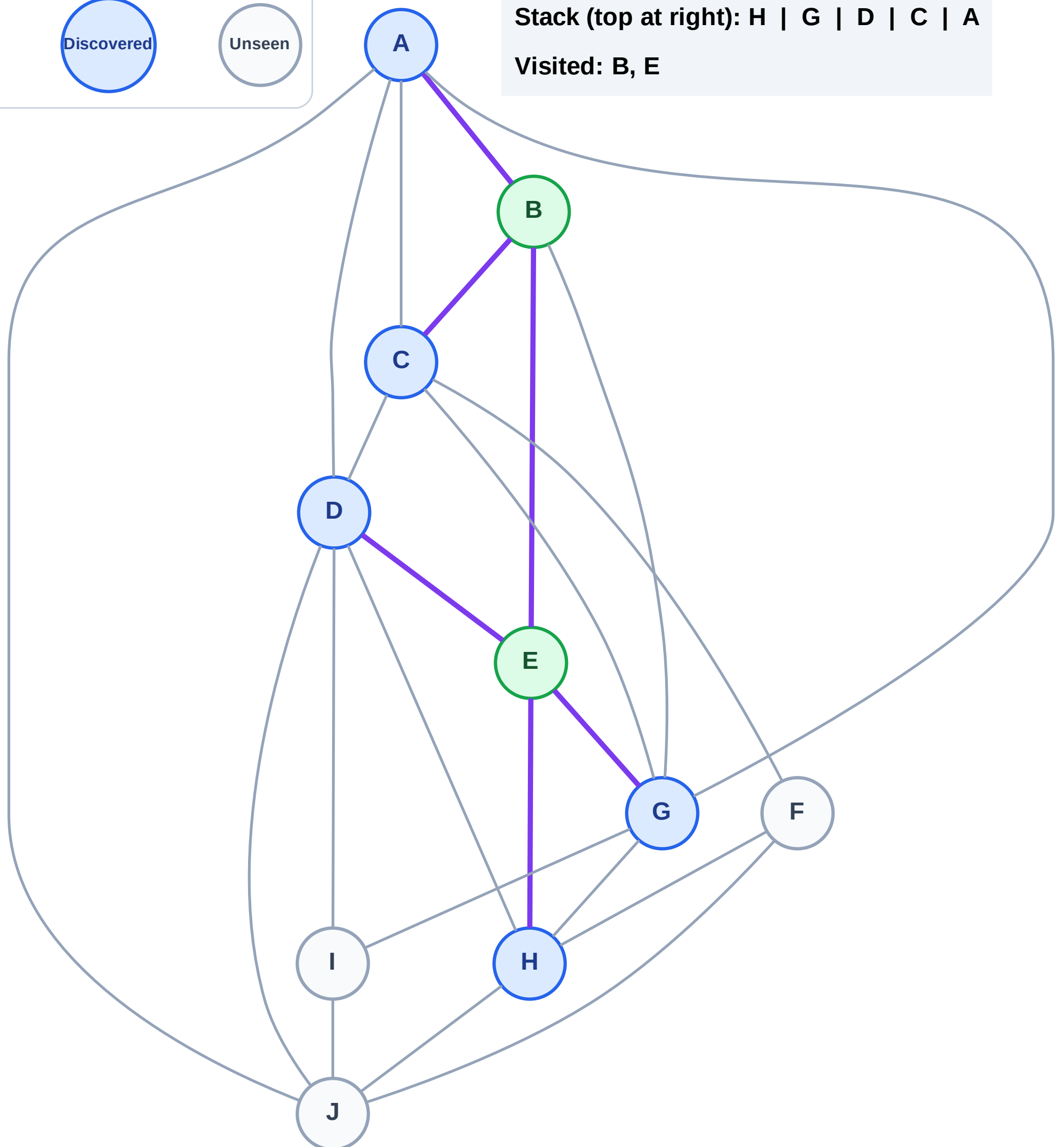
Step 5: Discover C, A from B

Legend



Stack (top at right): H | G | D | C | A

Visited: B, E



DFS Traversal

Step 6: Process node A

Legend

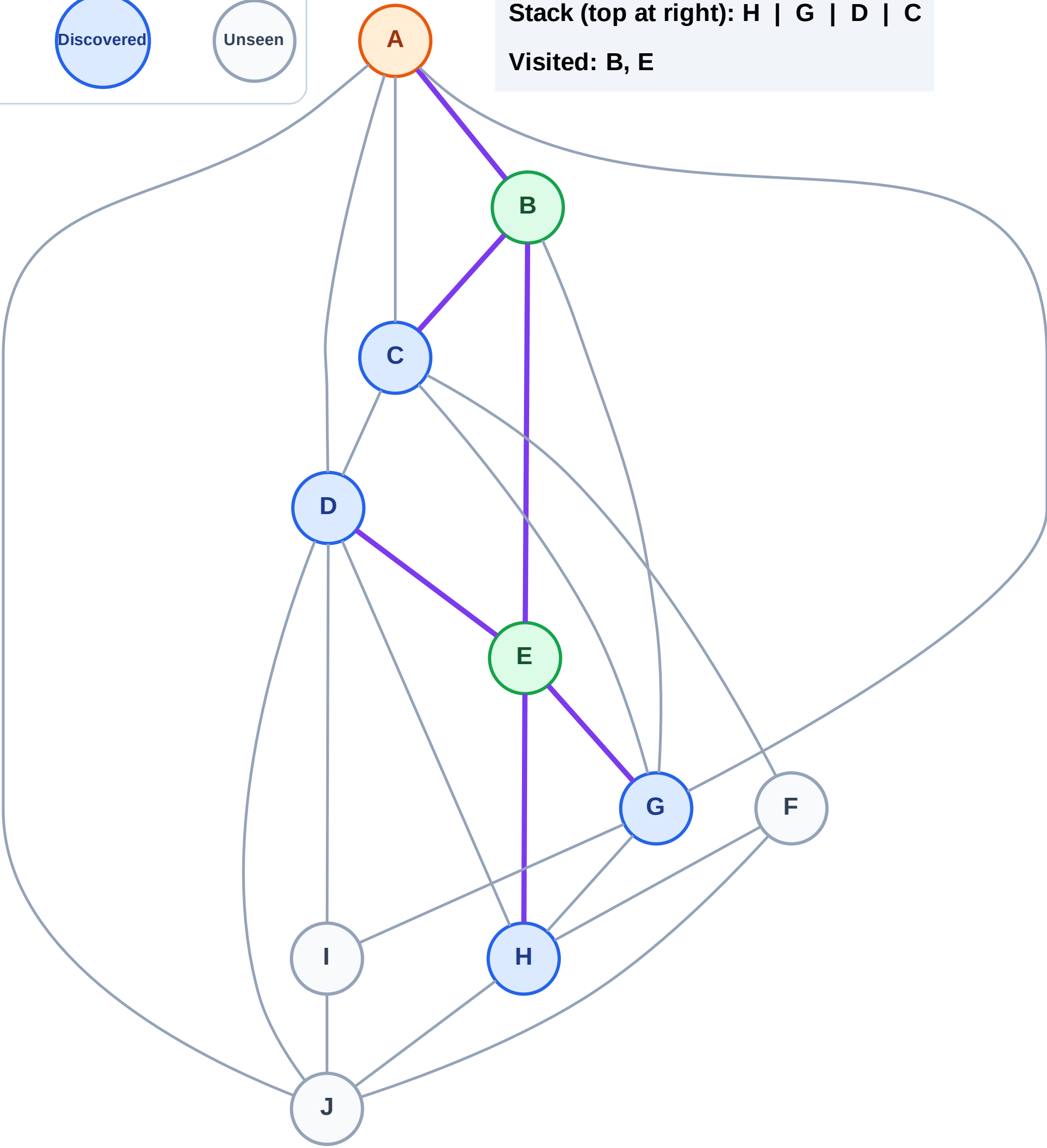
Complete

Processing

Discovered

Unseen

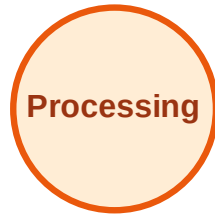
Stack (top at right): H | G | D | C
Visited: B, E



DFS Traversal

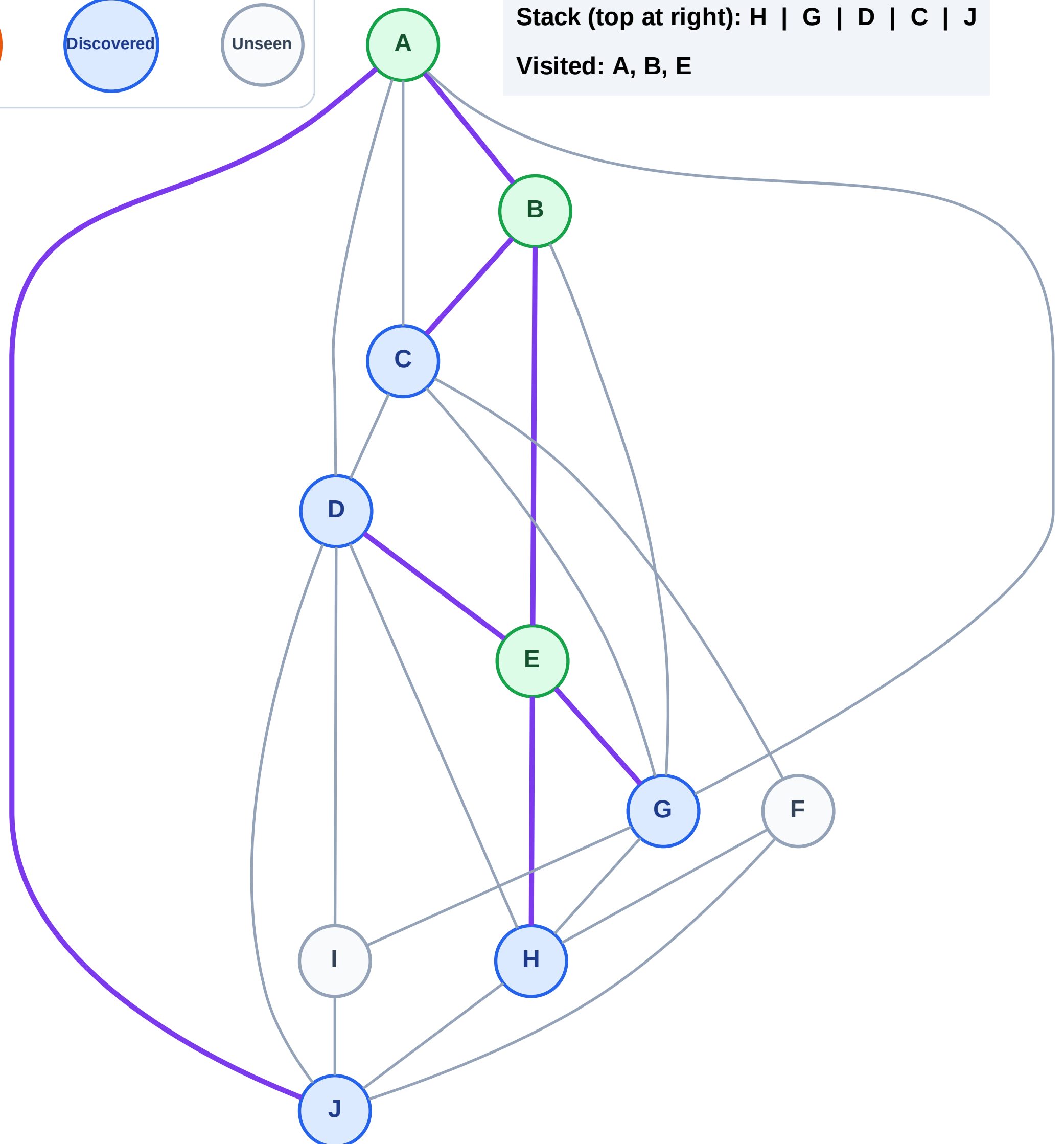
Step 7: Discover J from A

Legend



Stack (top at right): H | G | D | C | J

Visited: A, B, E



DFS Traversal

Step 8: Process node J

Legend

Complete

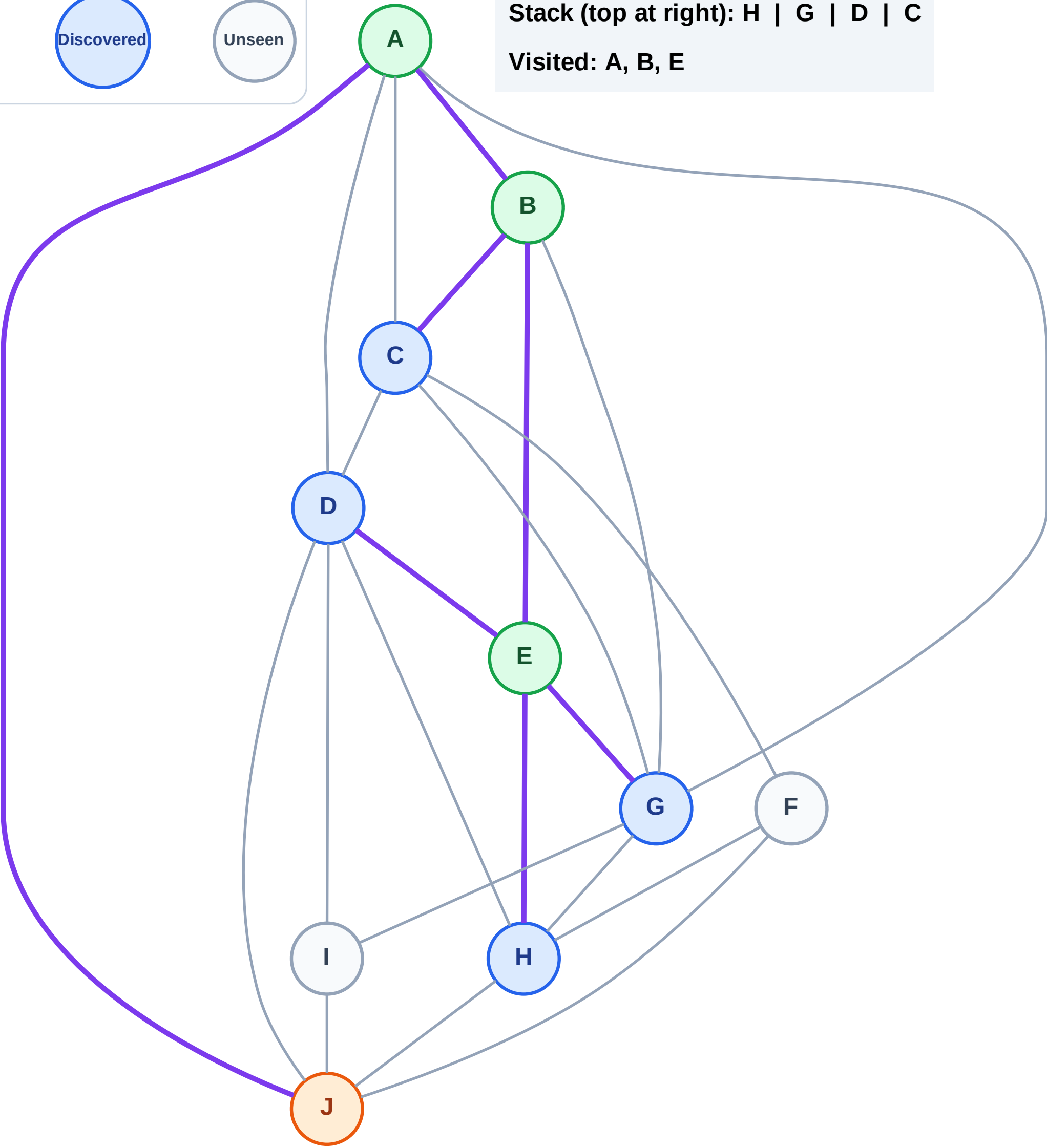
Processing

Discovered

Unseen

Stack (top at right): H | G | D | C

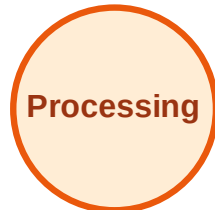
Visited: A, B, E



DFS Traversal

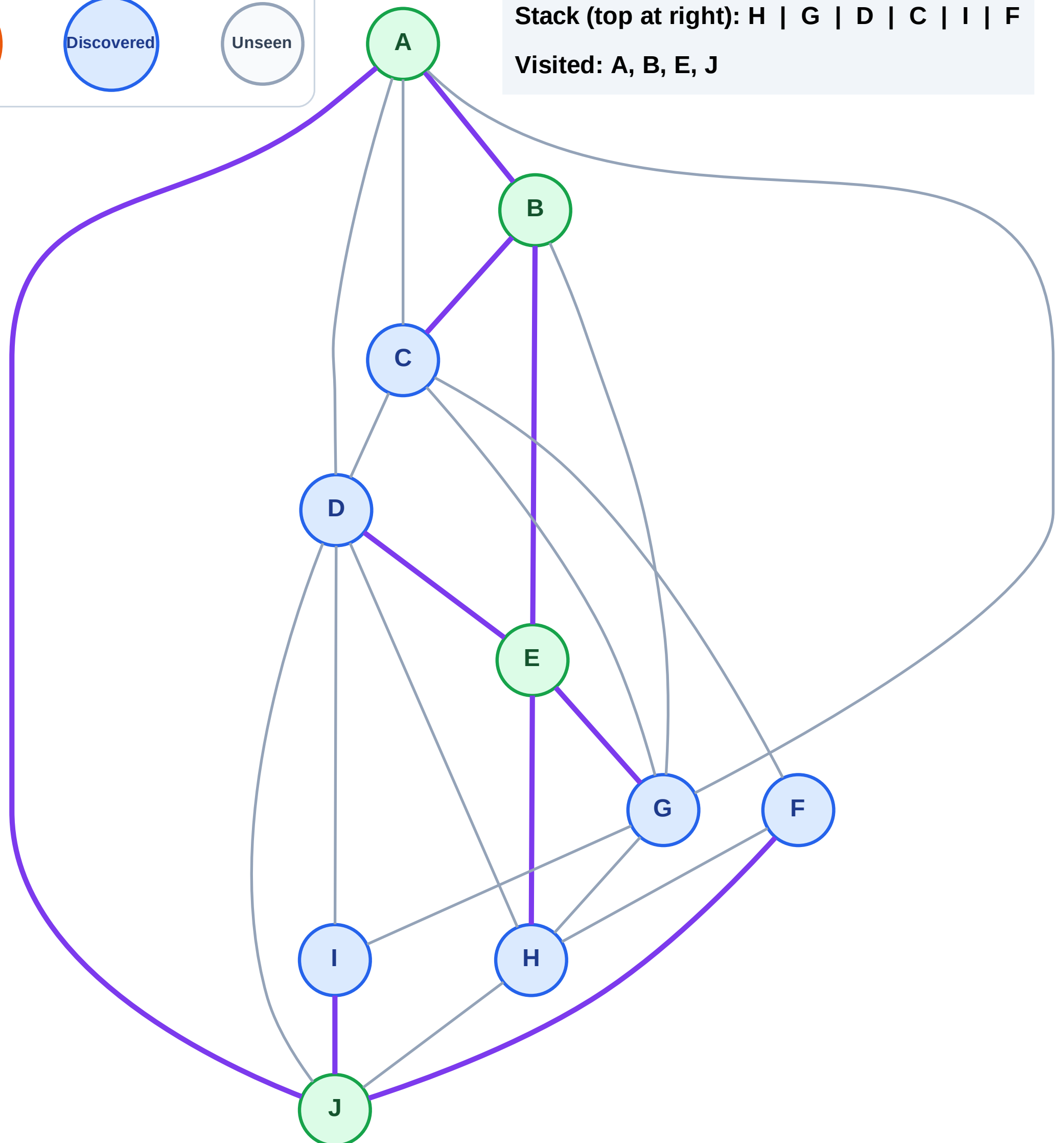
Step 9: Discover I, F from J

Legend



Stack (top at right): H | G | D | C | I | F

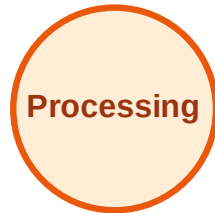
Visited: A, B, E, J



DFS Traversal

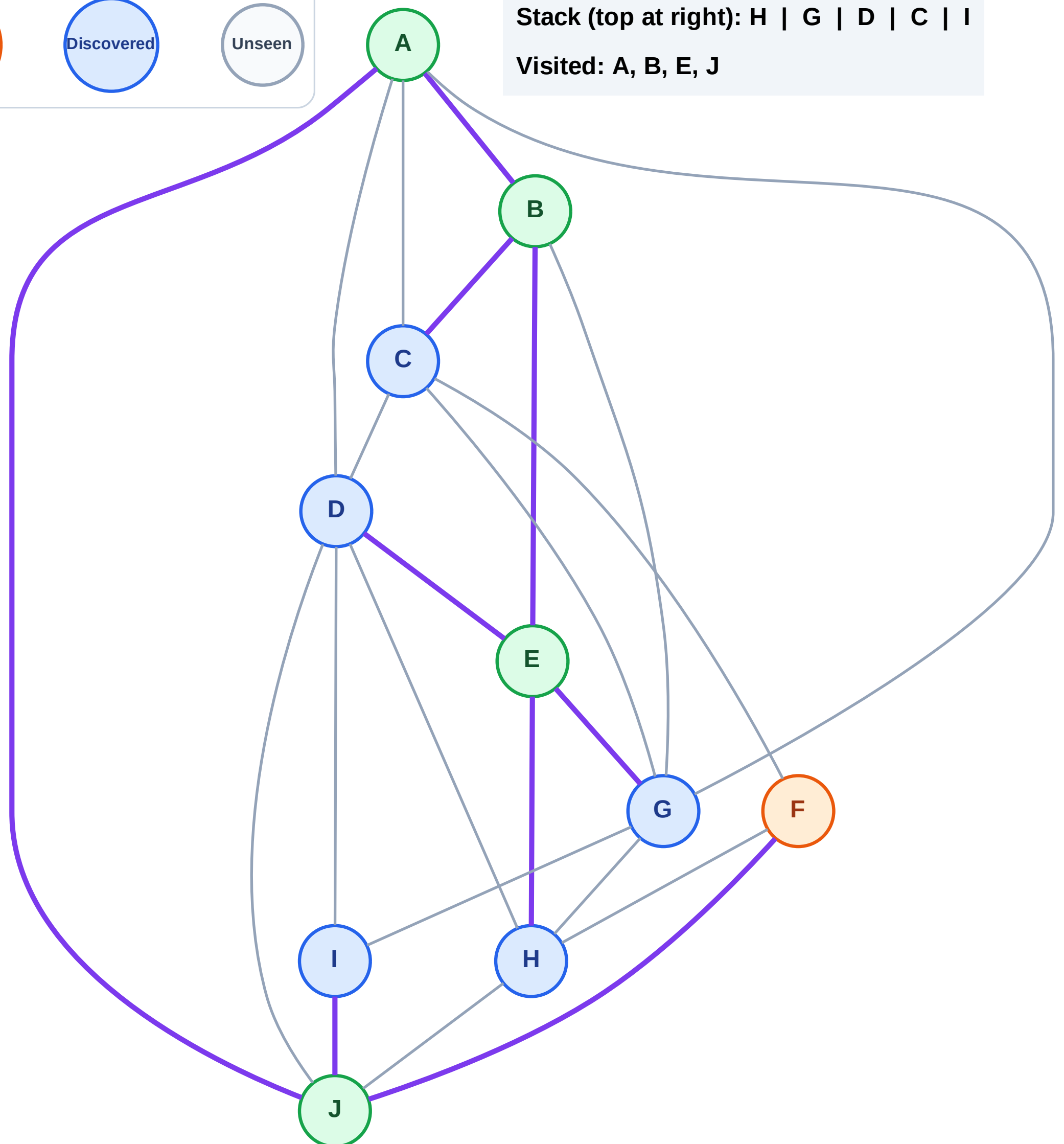
Step 10: Process node F

Legend



Stack (top at right): H | G | D | C | I

Visited: A, B, E, J



DFS Traversal

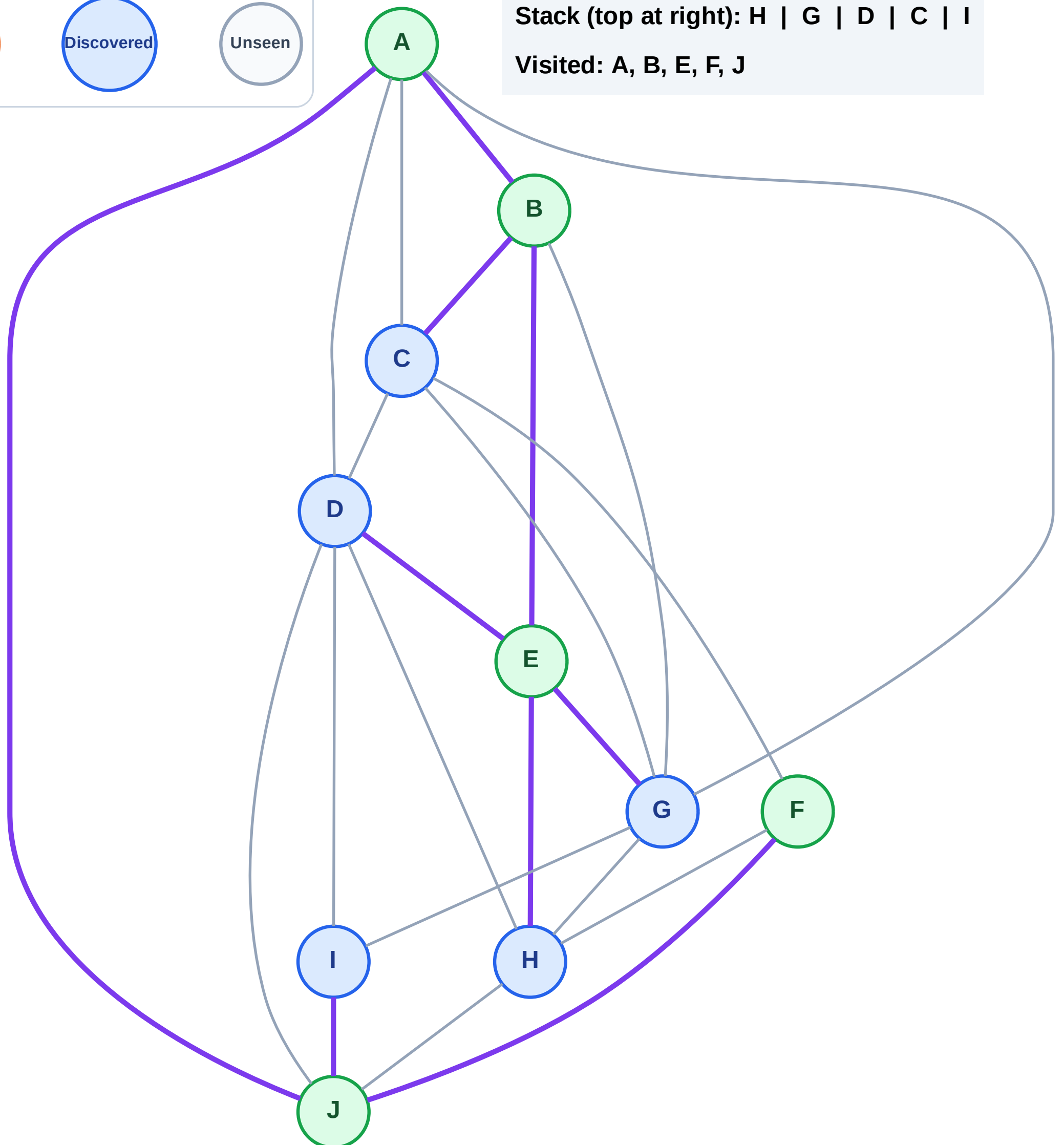
Step 11: No new neighbors from F

Legend



Stack (top at right): H | G | D | C | I

Visited: A, B, E, F, J



DFS Traversal

Step 12: Process node I

Legend

Complete

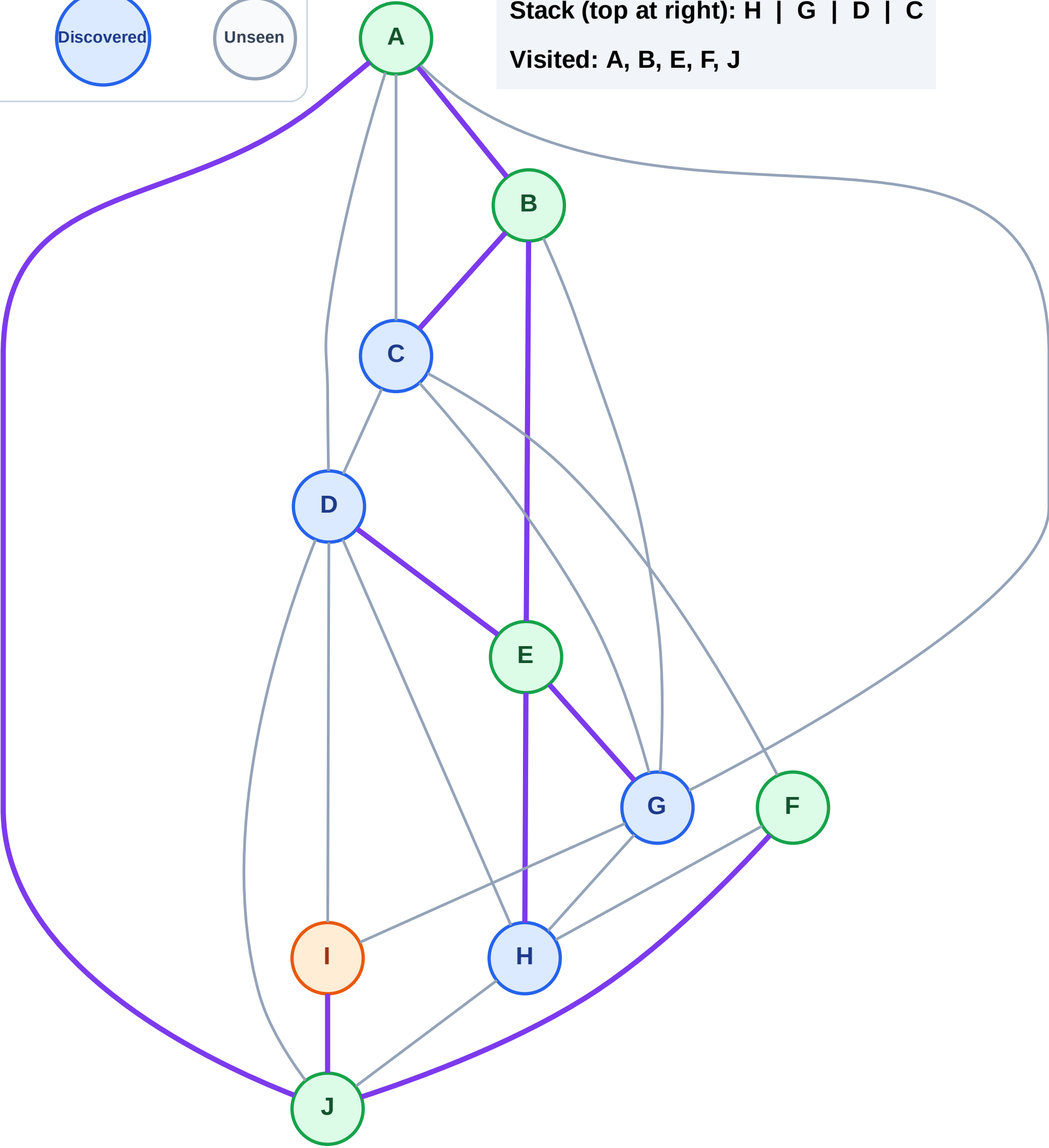
Processing

Discovered

Unseen

Stack (top at right): H | G | D | C

Visited: A, B, E, F, J



DFS Traversal

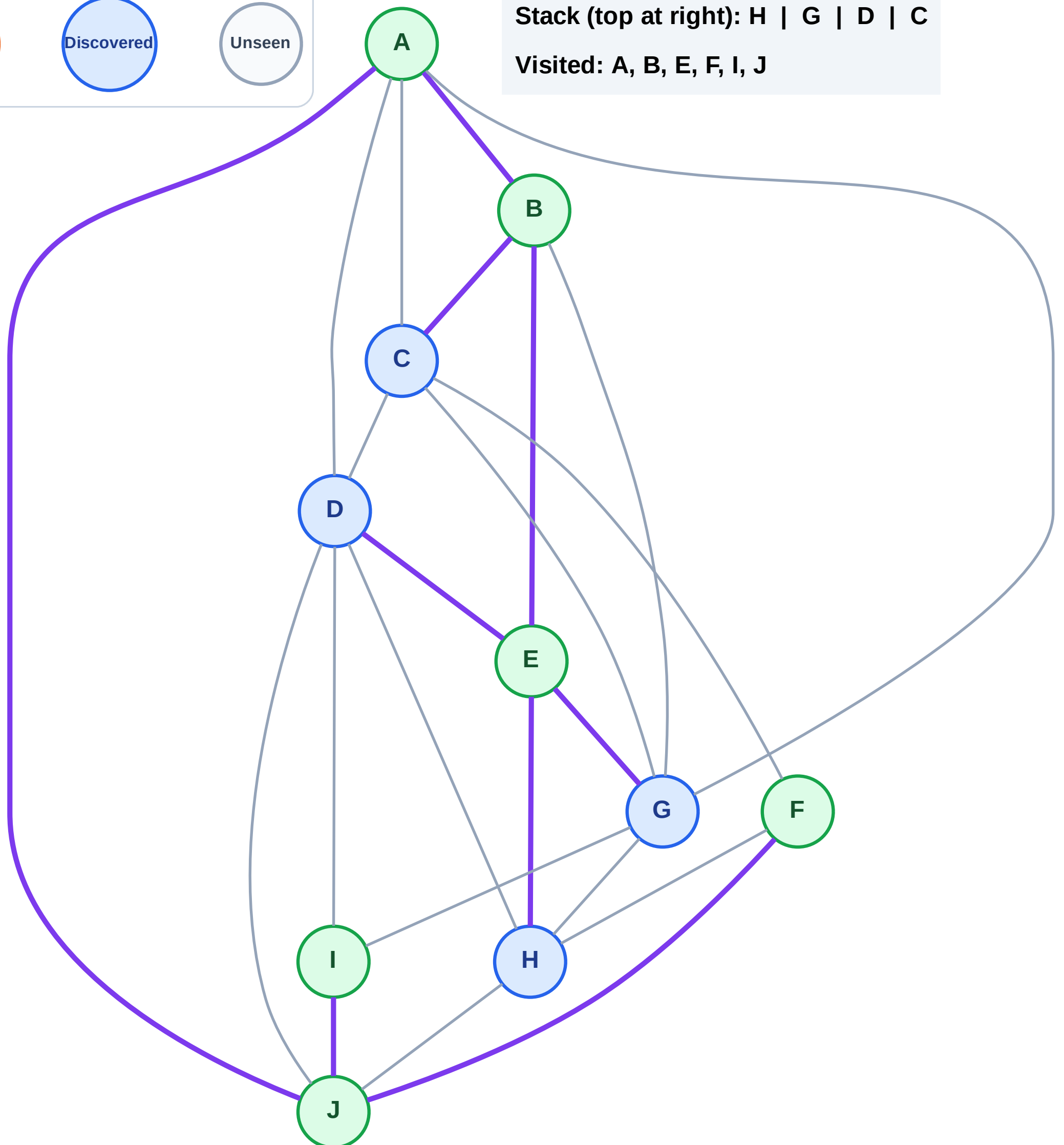
Step 13: No new neighbors from I

Legend



Stack (top at right): H | G | D | C

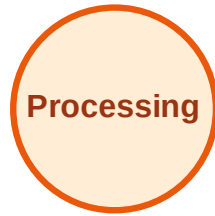
Visited: A, B, E, F, I, J



DFS Traversal

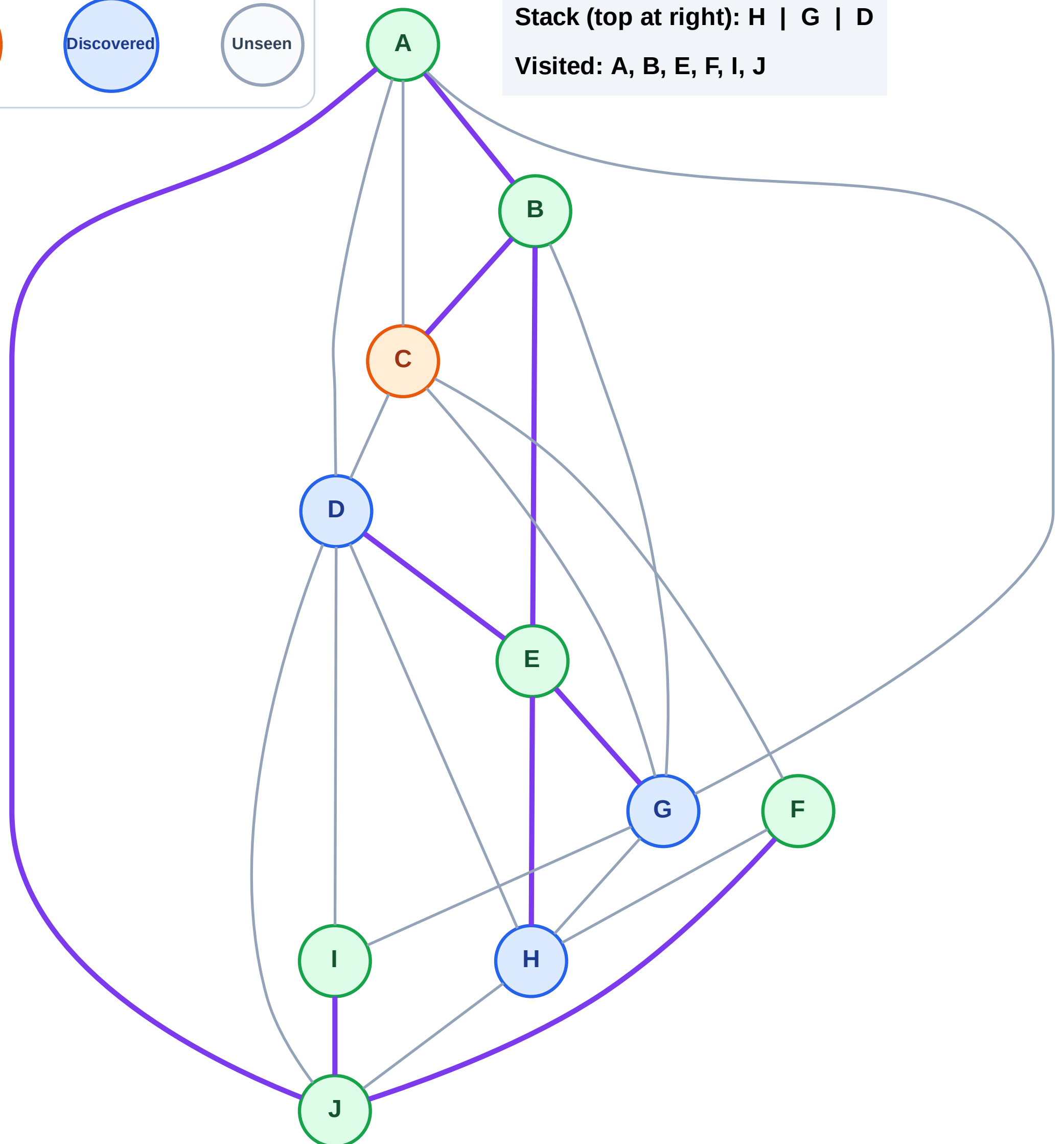
Step 14: Process node C

Legend



Stack (top at right): H | G | D

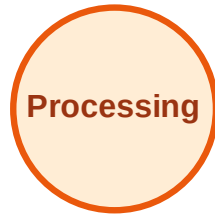
Visited: A, B, E, F, I, J



DFS Traversal

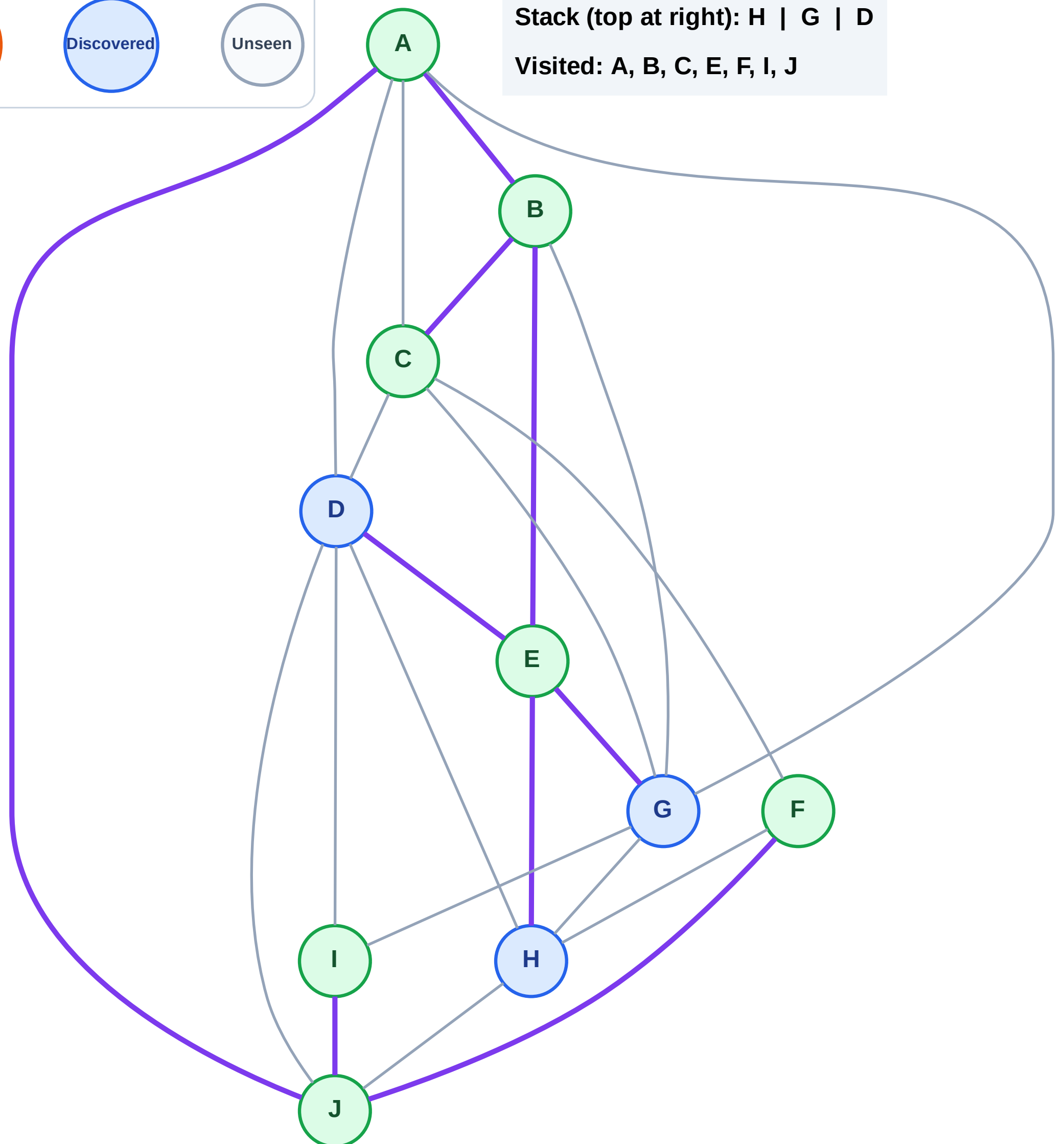
Step 15: No new neighbors from C

Legend



Stack (top at right): H | G | D

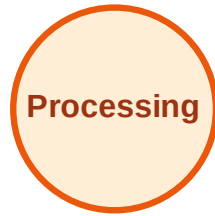
Visited: A, B, C, E, F, I, J



DFS Traversal

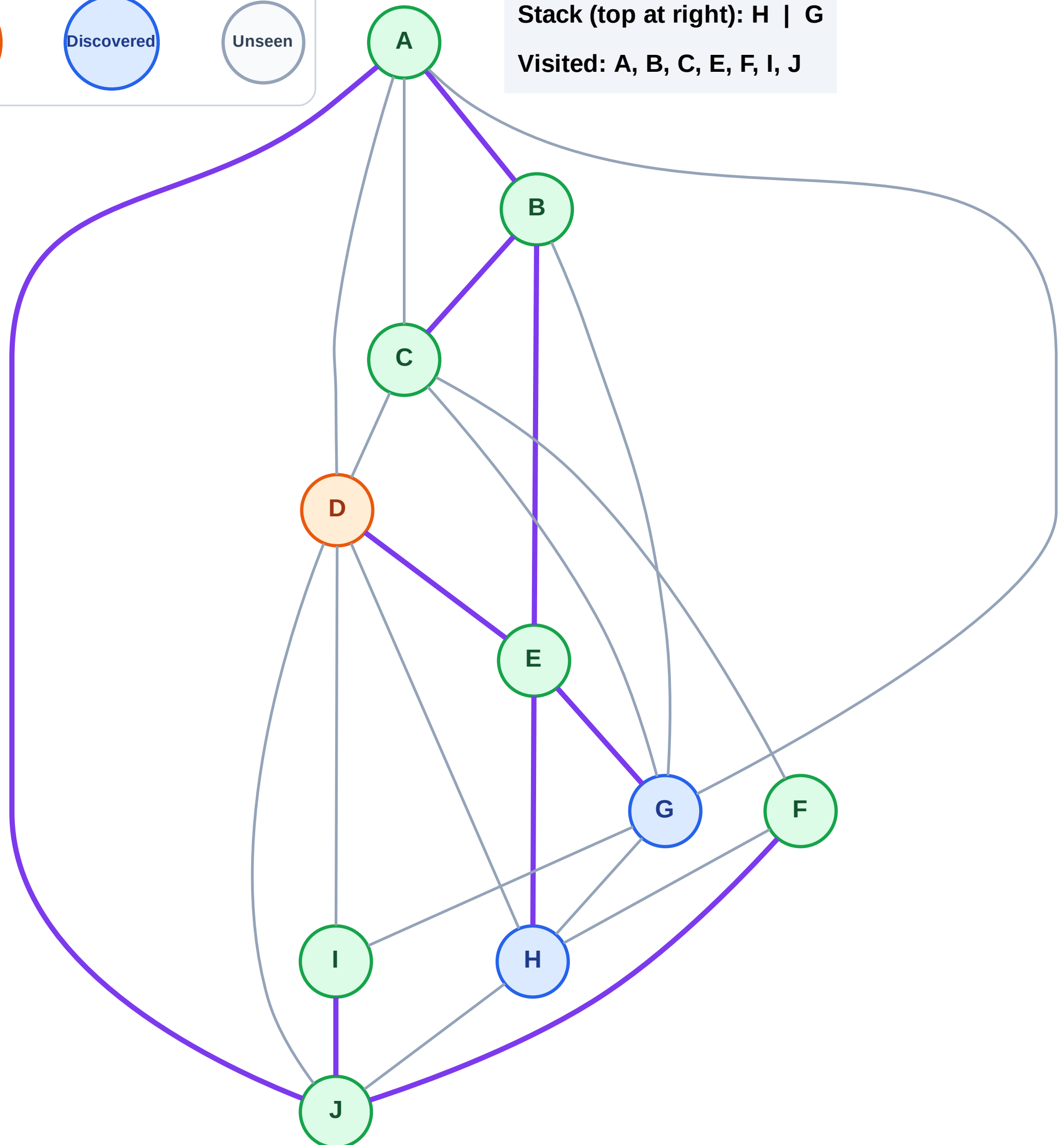
Step 16: Process node D

Legend



Stack (top at right): H | G

Visited: A, B, C, E, F, I, J



DFS Traversal

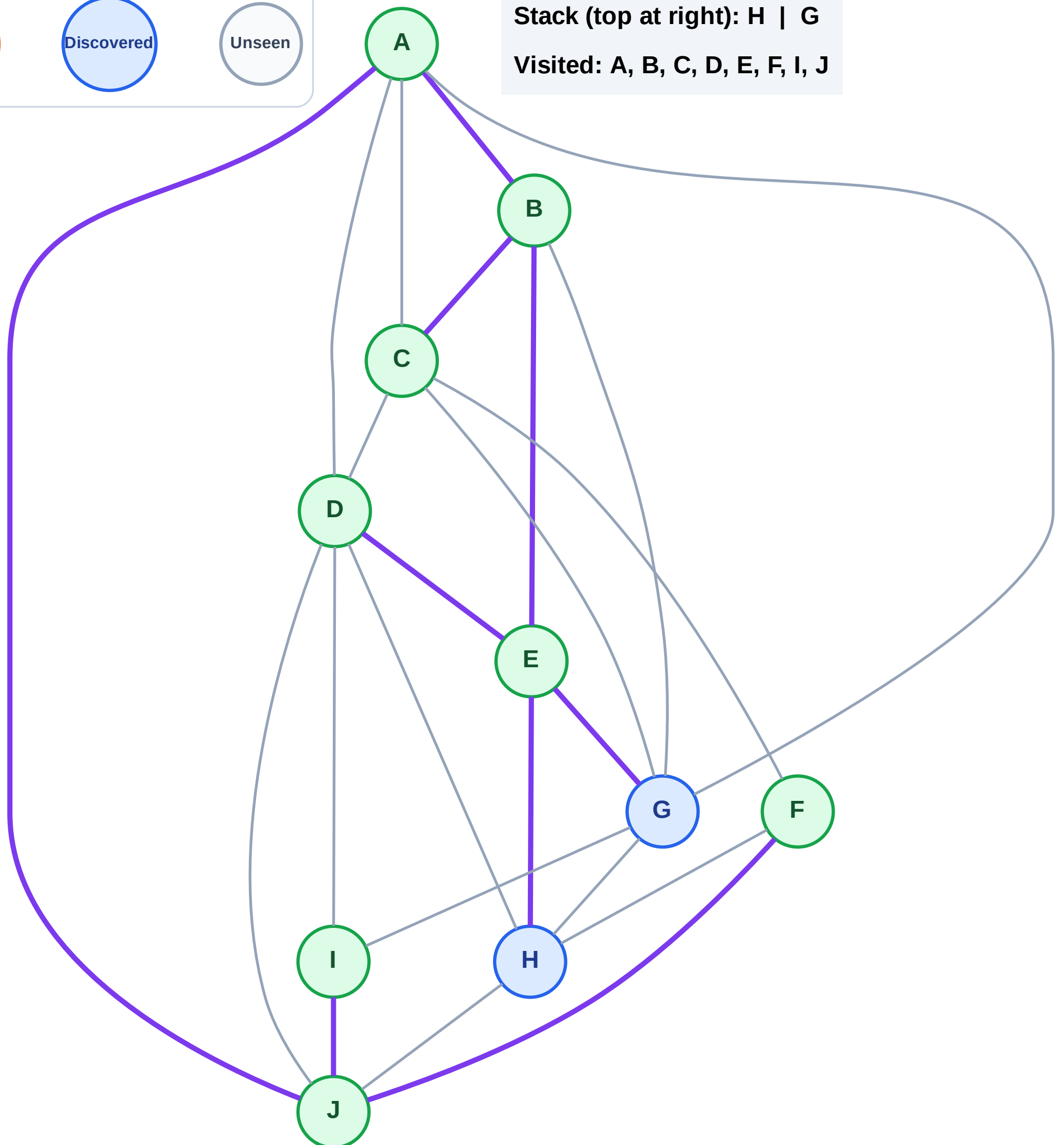
Step 17: No new neighbors from D

Legend



Stack (top at right): H | G

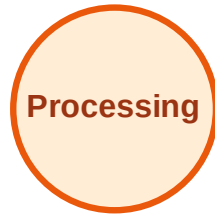
Visited: A, B, C, D, E, F, I, J



DFS Traversal

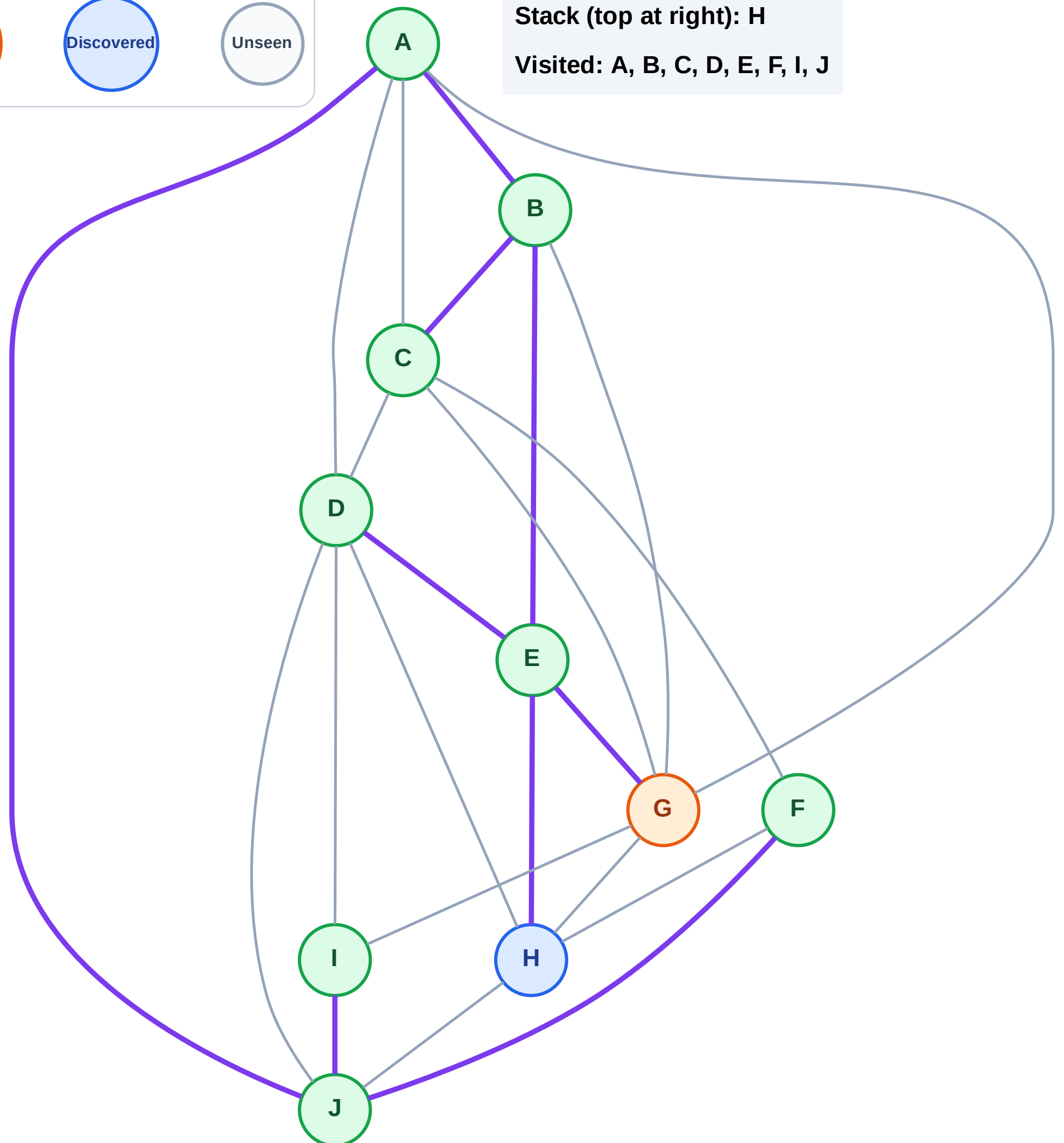
Step 18: Process node G

Legend



Stack (top at right): H

Visited: A, B, C, D, E, F, I, J



DFS Traversal

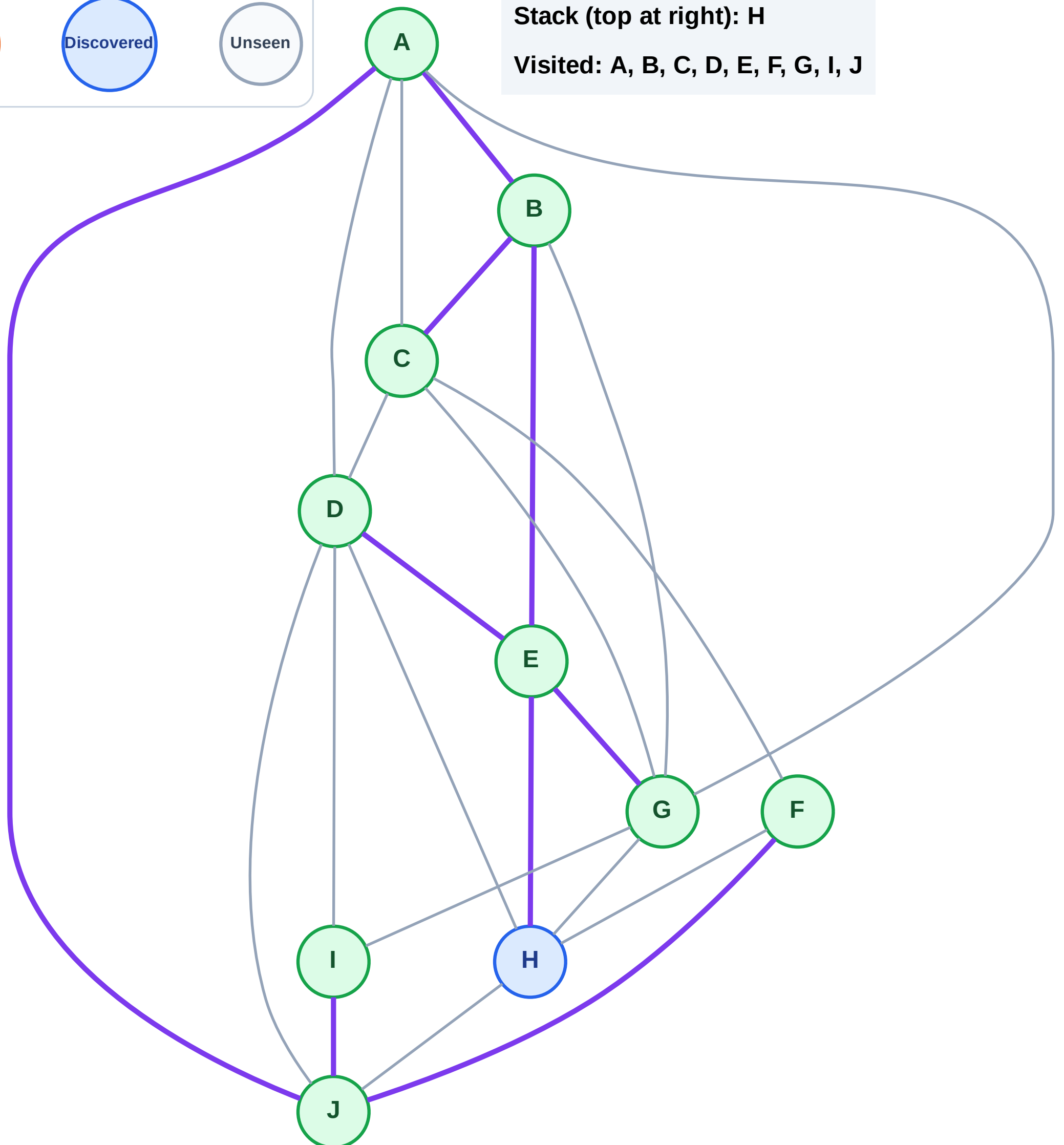
Step 19: No new neighbors from G

Legend



Stack (top at right): H

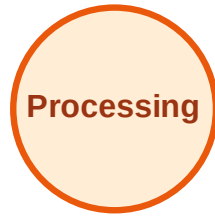
Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

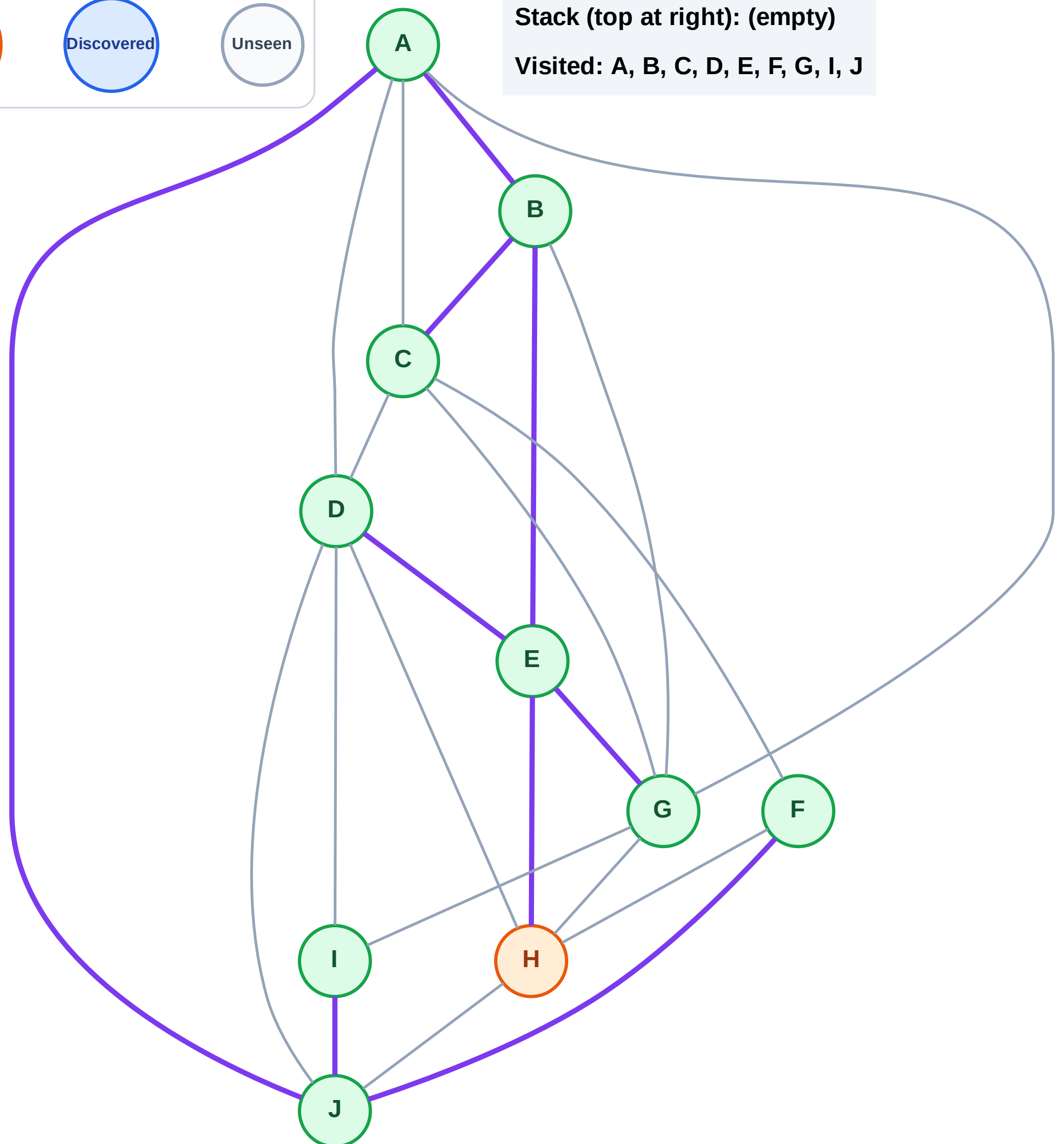
Step 20: Process node H

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, I, J



Step 21: No new neighbors from H

Step 21: No new neighbors from H

